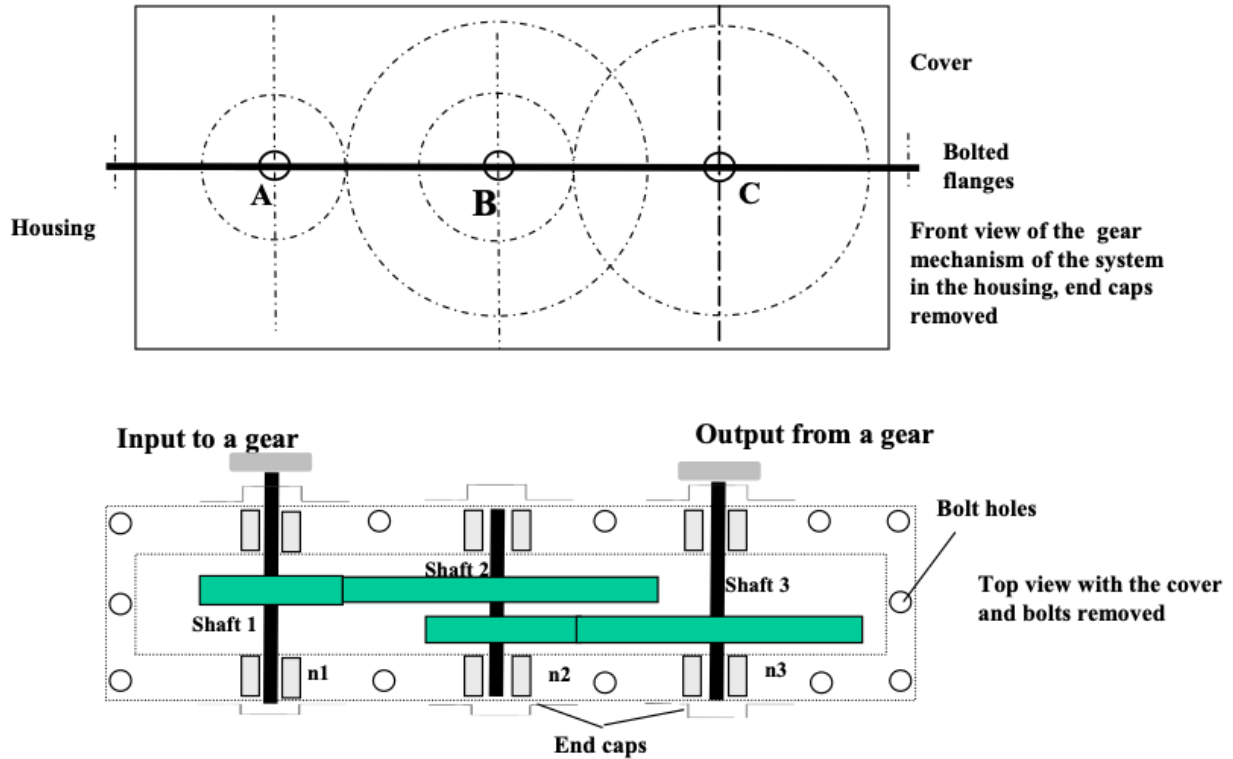


ME 315
THEORY OF MACHINES – DESIGN OF ELEMENTS
Fall 2024

Design Project
Group submission



Group members

Danish Galebotswe	Shaft system 1
Lochlan McGinnis	Shaft system 2
Daniel Lechuga	Shaft system 3

Grade

On-time step dues	
Structural design, drawings	/40
Design analysis	/35
Report writing	/20

Total _____/95

Introduction

This design report presents the analysis and development of a three-shaft gear train system for transmitting power from an input shaft (Shaft A) to an output shaft (Shaft C) with a speed reduction. The system is designed to achieve the following key specifications:

1. **Power and Speed:** The motor speed is 2700 rpm with a power input of 60 kW, and the output shaft speed (n_3) is to be maintained at 400 ± 3 rpm.
2. **Efficiency and Reliability:** Assuming 100% efficiency for each gear pair and targeting a reliability of 99%, the design is expected to operate continuously for at least 5 years under the specified conditions.
3. **Structural Considerations:** The design requires determining the optimal distance between shafts A and C, ensuring a minimum shaft diameter of 20 mm for stiffness, and considering fatigue stress concentration factors for bending and torsional stresses.

The report details the design process for each of the three shafts, focusing on achieving a compact, cost-effective, and efficient structure. Each team member is responsible for the design and analysis of one shaft, including gear, shaft, and bearing selections supported by detailed calculations. Additional components such as spacers, seals, and end caps are selected based on geometric design considerations.

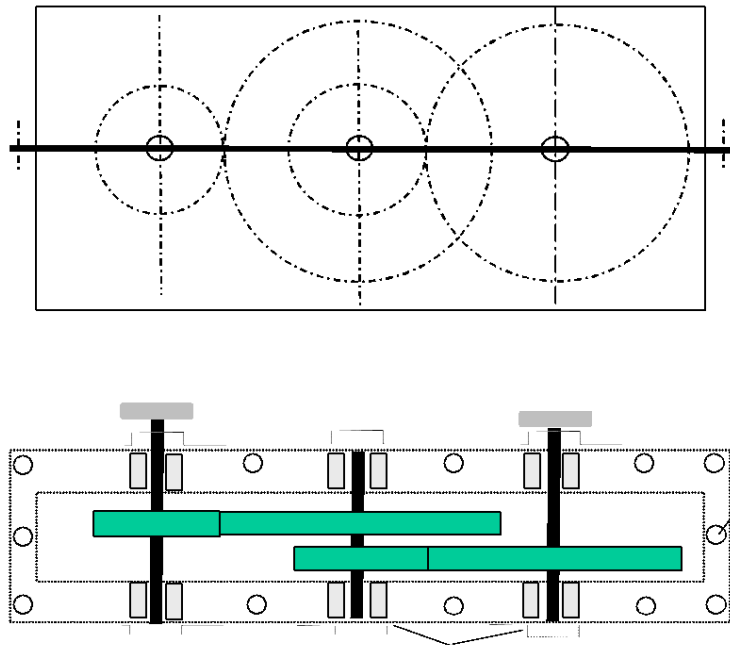


Figure 1: Schematic of the Transmission.

Scope of the Design

The scope includes the following aspects:

- Gear selection and arrangement (spur or helical gears).
- Force, geometry, and fatigue calculations necessary for shafts, bearings and gears.
- Compact layout of the gear train within a housing separated at the plane of shaft centerlines.
- Consideration of fits and tolerances for gear and bearing mountings.

Design Philosophy

While the housing design is not required, a rough housing design will be considered for conceptual convenience. The team aims to maximize performance and compactness while minimizing costs, guided by a structured approach to ensure durability and reliability in real-world operating conditions.

This report documents the decisions, methodologies, and calculations that underpin the design of the gear train system, demonstrating a robust and reliable solution for the power transmission application.

Gear Train and gear parameters

This section provides a comprehensive summary of the key parameters and design considerations for the three-shaft gear train system. The details include the power, speed, and geometry of each gear, as well as the forces, materials, and safety factors involved in the design. Additionally, the centerline distances between shafts and the total distance between Shafts A and C are specified.

The information is presented in Table 1, is structured to include:

- Power and speed (rpm) of each gear.
- Transmission error.
- Major geometric properties of the gears, including module, number of teeth, pressure angle, and face width.
- Centerline distances between shafts and the total centerline distance AC.
- Torque and gear forces.
- Material properties, hardness, and safety factors.

Note that: it is assumed that there is no power lost due to friction/sound/heat.

Table 1: Geartrain and Gear Parameters

Item	Gear 1	Gear 2	Gear 3	Gear 4
Gear Type	Spur	Spur	Spur	Spur
Module (mm)	5	5	5	5
Pressure angle (deg)	20	20	20	20
Centerline Distance (mm)	195		210	
Total Centerline Distance (mm)	405			
Speed (rpm)	2700	1060.71	1060.71	399.94
Resultant Transmission Error	$\frac{399.94 - 400}{400} = 0.01\%$			
Power (kW)	60	60	60	60
Number of teeth (N)	22	56	23	61
Addendum (mm)	5	5	5	5

Dedendum (mm)	6	6	6	6
Tooth height (mm)	11	11	11	11
Clearance (mm)	1	1	1	1
Circular Pitch (mm)	16	16	16	16
Base Pitch (mm)	15	15	15	15
Base width (mm)	65	60	65	60
Tooth Thickness (mm)	8	8	8	8
Pitch circle diameter (mm)	110	280	115	305
Outside circle diameter (mm)	120	290	125	315
Base circle diameter (mm)	103	263	108	287
Root circle diameter (mm)	97.5	267.5	102.5	292.5
Torque (Nm)	212.21	540.16	540.16	1432.60
Wt (N)	3858.30	3858.30	9394.13	9394.13
Wr (N)	1404.31	1404.31	3419.18	3419.18
Material	AISI 4140 Steel, normalized, quenched, tempered	AISI 1045 Steel, Quenched and Tempered to 450 HB	AISI 4140 Steel, normalized, quenched, tempered	AISI 1045 Steel, Quenched and Tempered to 450 HB
Hardness	578	450	578	450
Bending Factor of Safety	5.85	4.56	2.45	2.17
Contact Factor of Safety	2.04	1.55	1.43	1.15

This table summarizes the critical design outputs and serves as a quick reference for evaluating the performance and feasibility of the gear train. Detailed calculations supporting the values in the table are included in the Appendix for further reference and validation.

Shafts and Parameters

Table 2 provides a comprehensive overview of the design and operational parameters for shafts used in mechanical systems. The table includes key details such as the material properties, rotational speeds, maximum torque and bending moments, and the location of critical cross-sections. Additionally, it highlights the factors of safety applied, and whether the expected life of the shafts is considered infinite or finite. This information is critical for ensuring the shafts meet performance requirements under specified loading and operating conditions.

Material: AISI 1020 Steel, as rolled
 Ultimate Tensile Strength: 420 MPa
 Yield Strength: 350 MPa

Table 2: Shafts and Parameters

Item	Shaft 1	Shaft 2	Shaft 3
Speed (rpm)	2700	1060.71	399.94
Maximum torque (Nm)	212.21	540.16	1432.60
Maximum bending moment (Nm)	127	235.04	377.69
Location of critical cross section	At Gear 1 ⁺ ($x = 130.5 \text{ mm}$) Where torque is applied	At Gear 3 ⁺ ($x = 58 \text{ mm}$) Where torque is applied	At Gear 4 ⁺ ($x = 58 \text{ mm}$) Where torque is applied
Materials	AISI 1020 Steel, as rolled	AISI 1020 Steel, as rolled	AISI 1020 Steel, as rolled
Goodman Factor of Safety	2.35	4.39	1.38
Yield Factor of Safety	5.99	7.67	2.18
Expected Life	Infinite	Infinite	Infinite

Bearings and parameters

The bearing type selected is SKF Deep Groove Ball Bearings due to low friction and therefore being ideal for high speed application. Another reason for selecting deep groove ball bearings is due to the low thrust load through each shaft due to using spur gears (zero thrust load for all the bearings). Table 3 details the specifications and parameters of bearings used in the system, providing essential information for selection and performance evaluation. The table includes data on radial and thrust loads, bearing identification numbers, inner and outer diameters, widths, dynamic and static load ratings (C and Co), and the estimated service life for each bearing set. Additionally, it indicates whether bearing-change services are anticipated, ensuring clarity in maintenance and operational planning.

Table 3: Bearings and Parameters

Item	Shaft 1		Shaft 2		Shaft 3	
Bearing:	Bearing A	Bearing B	Bearing C	Bearing D	Bearing E	Bearing
Radial Load (N)	15583.36	46780.24	58477.01	79015.79	45864.43	52079.02
Thrust Load (N)	0	0	0	0	0	0
Bearing Type	Deep Groove Ball Bearings	Deep Groove Ball Bearings	Deep Groove Ball Bearings	Deep Groove Ball Bearings	Deep Groove Ball Bearings	Deep Groove Ball Bearings
Bearing Numbers	SKF 6309	SKF 6309	SKF 6410	SKF 6410	SKF 6309	SKF 6309
ID (mm)	45	45	50	50	45	45
OD (mm)	100	100	130	130	100	100
Width (mm)	25	25	31	31	25	25
C (N)	52,700	52,700	81,700	81,700	52,700	52,700
Co (N)	31,500	31,500	52,000	52,000	31,500	31,500
Life (Mc)	2358.72	2358.72	926.64	926.64	349.39	349.39
Bearing Change Service	No	No	No	No	No	No

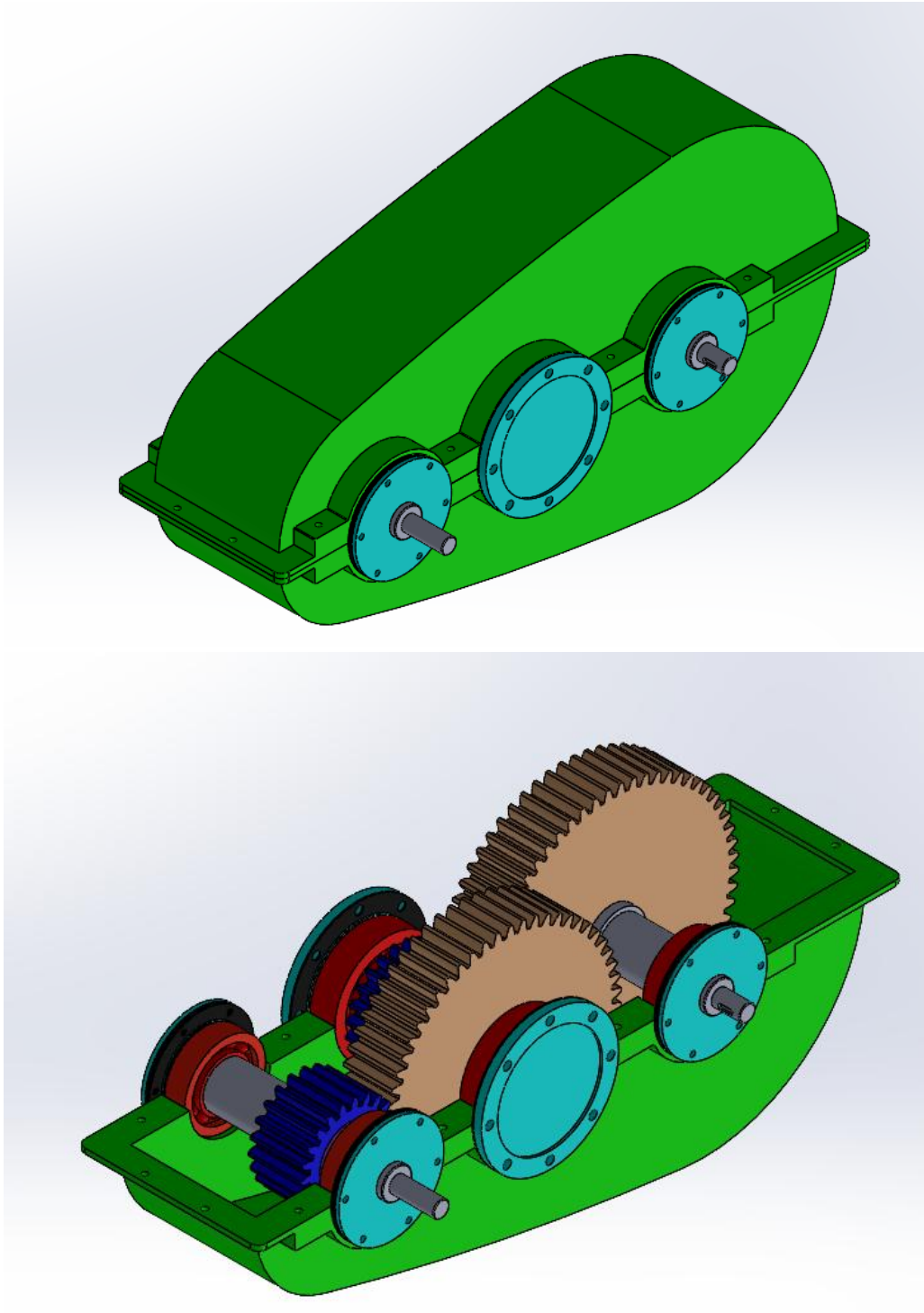
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Overall Results

Table 4: Overall Results

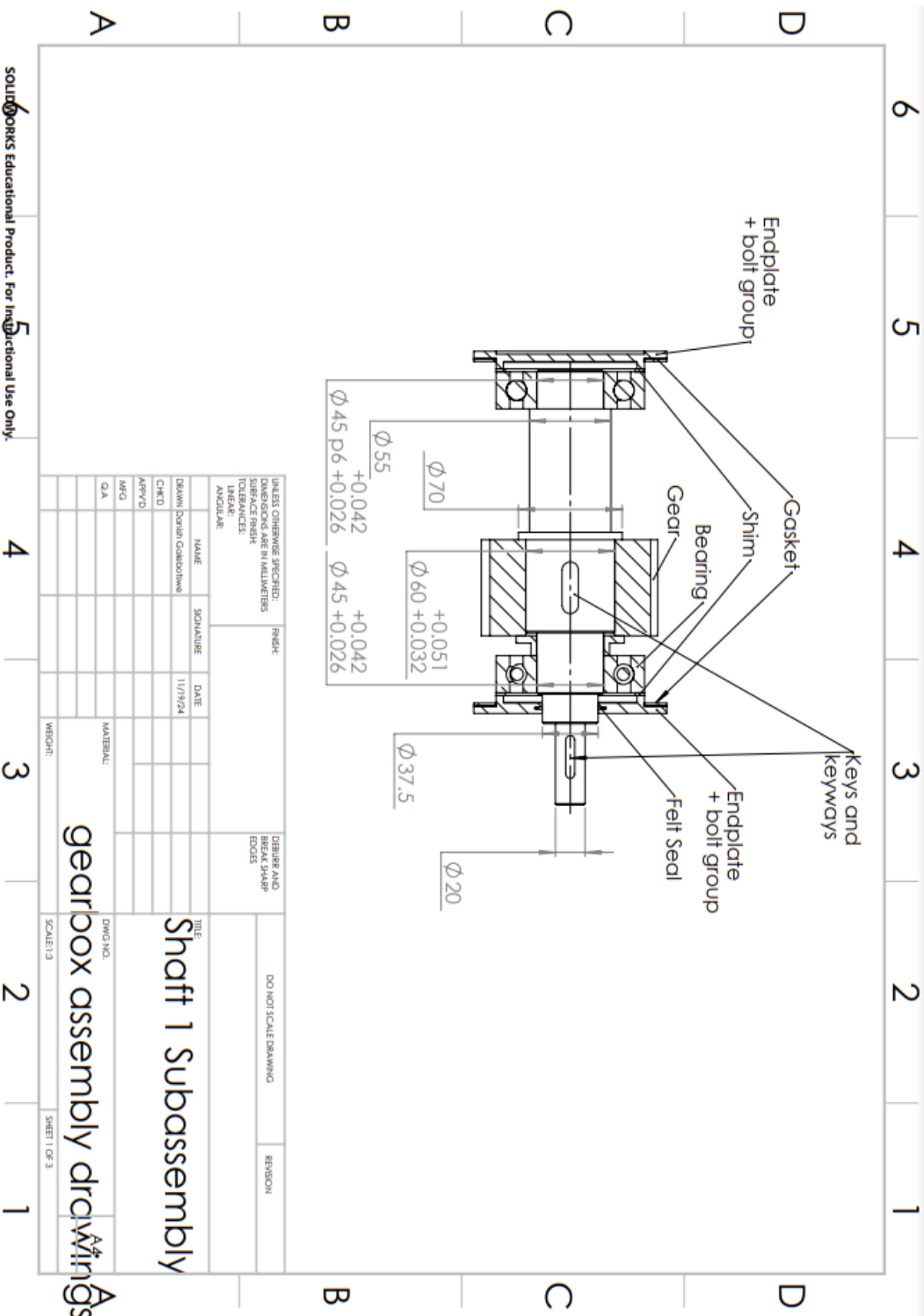
Item:	Value:
Power (kW)	60
Input speed (rpm)	2700
Output speed (rpm)	399.94
Overall Design Factor of Safety	1.15
Bearing Changes	The bearing life is calculated for 5 years, so no bearing change service is needed within this period. Maintenance operations should focus on monitoring for wear beyond the expected life.
Oil Changes	Oil changes are recommended annually to ensure proper lubrication and prevent wear or failure due to contamination. A felt seal is used to prevent oil leakage or debris ingress.
Additional Considerations	Bearings are standardized on the input and output shafts of the gearbox for ease of maintenance, reducing cost and assembly complexity. A bearing puller can easily remove bearings if required.

General Arrangement

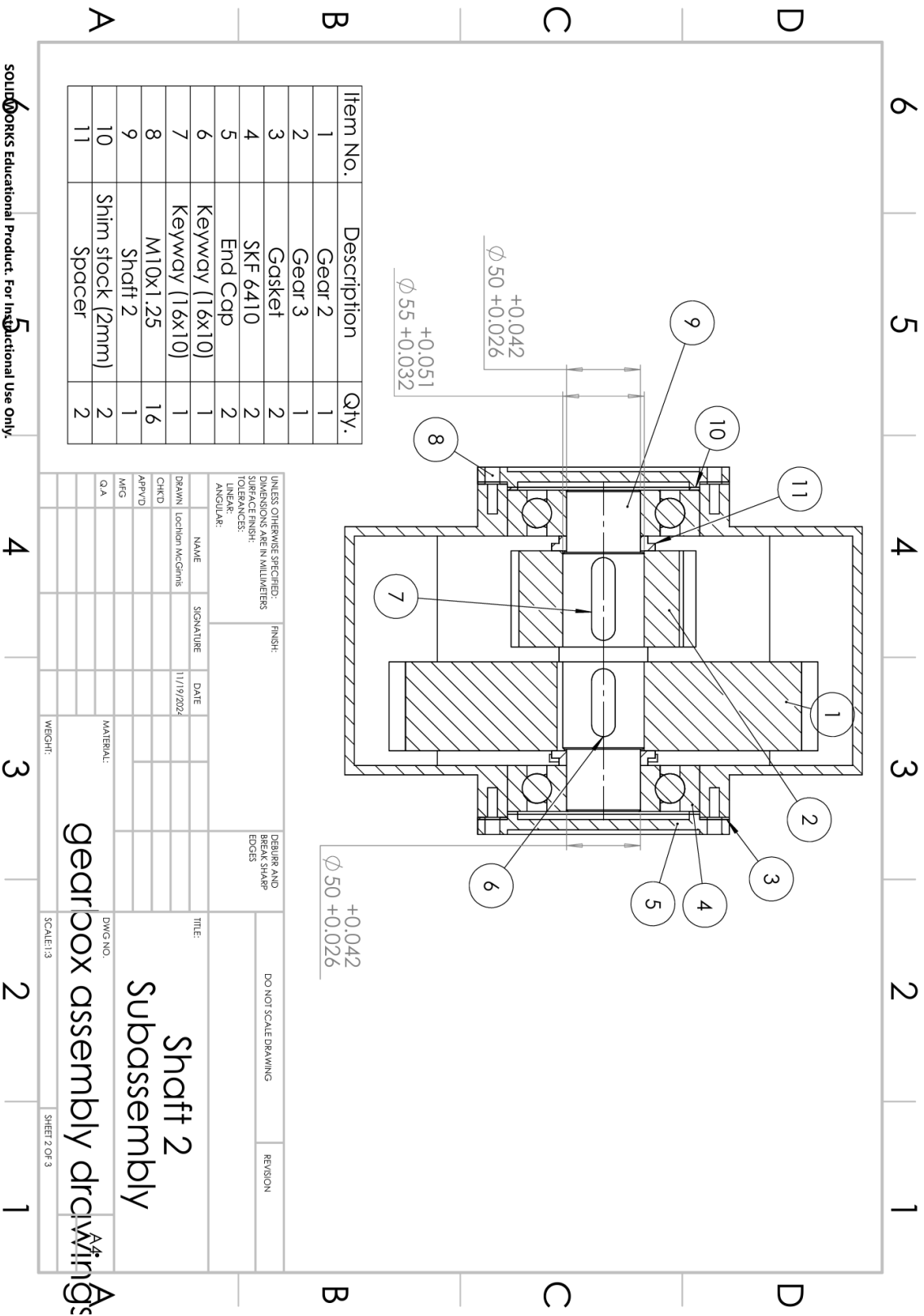


Shaft Assembly Drawings

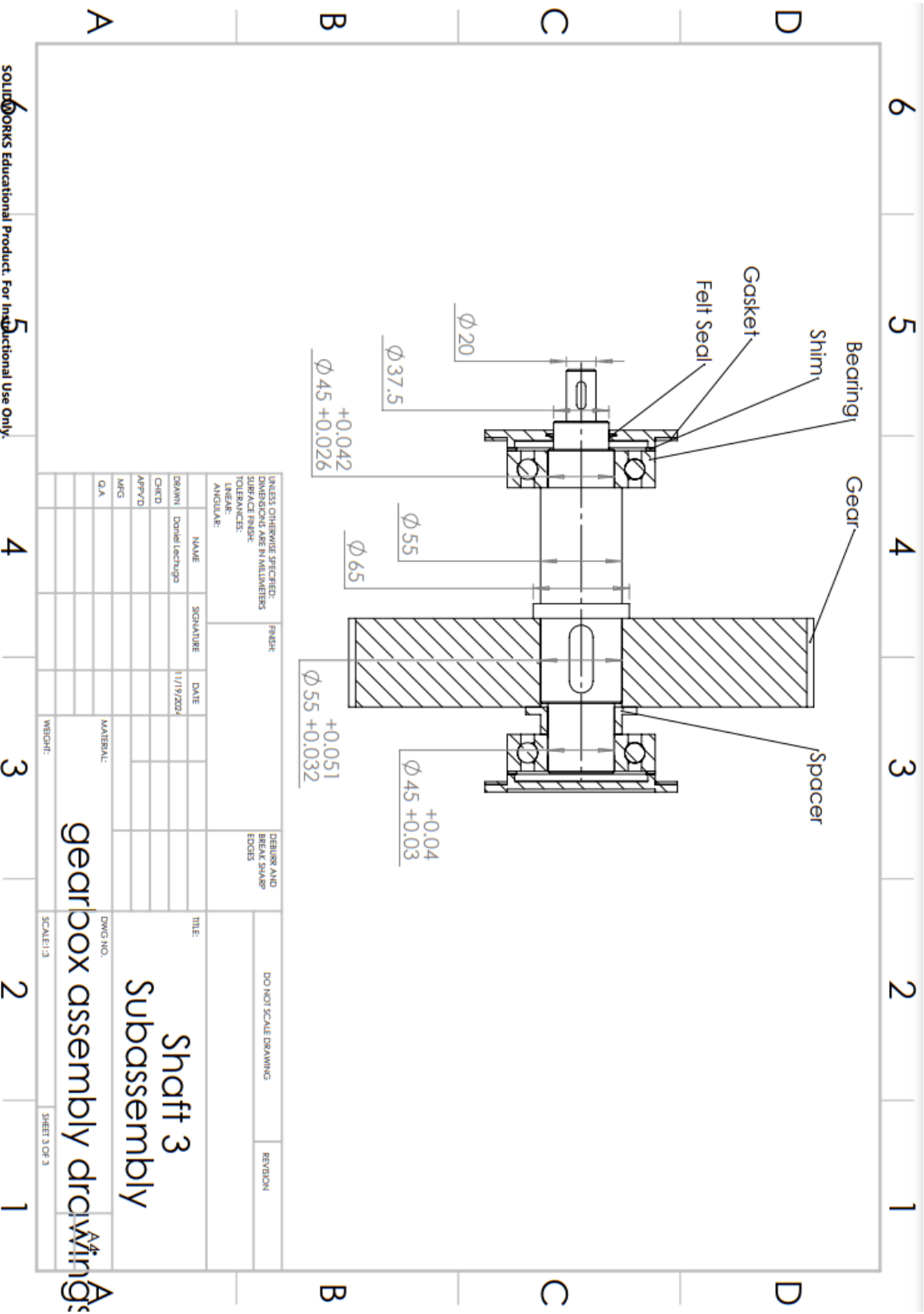
Shaft 1 Assembly



Shaft 2 Assembly

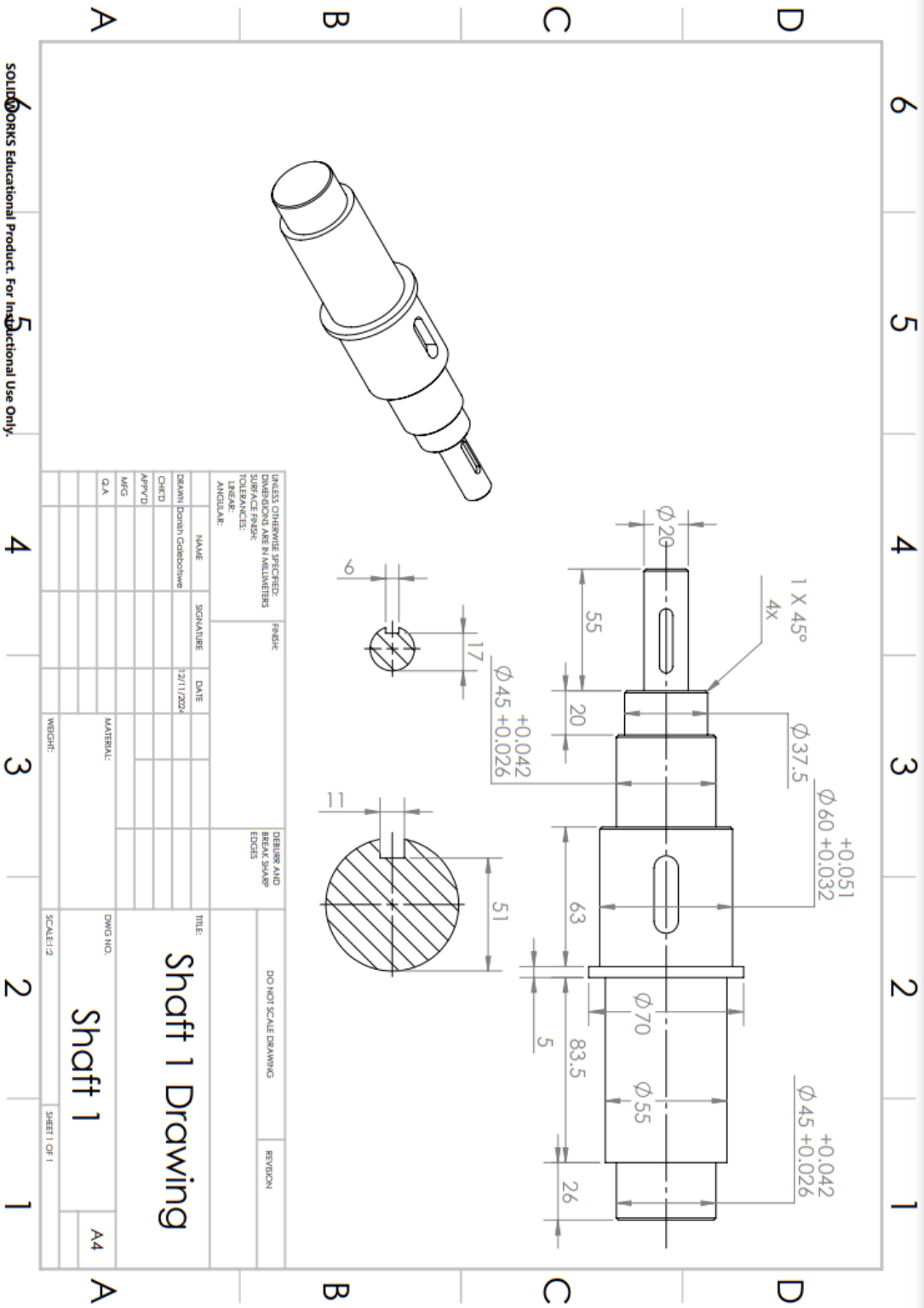


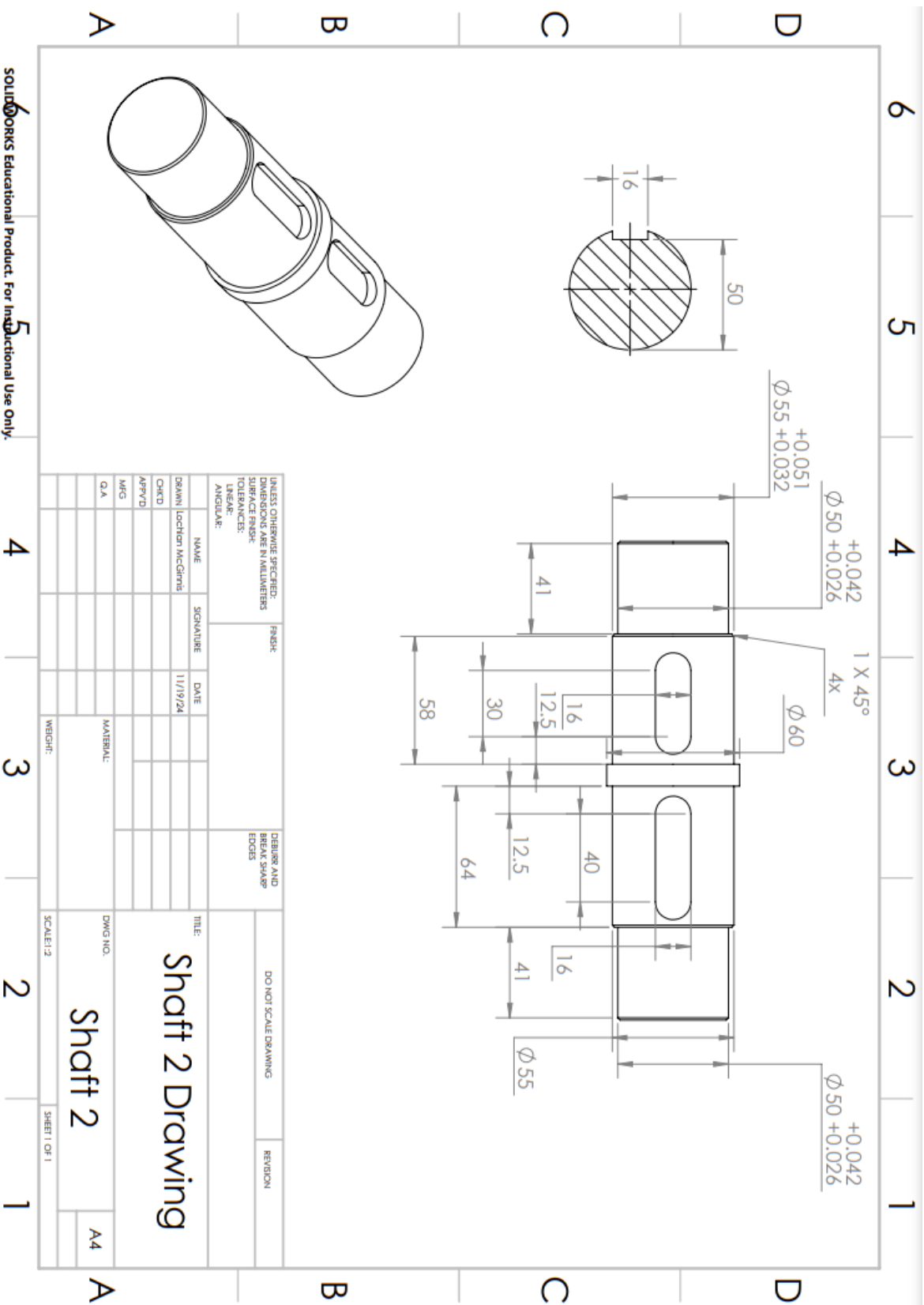
Shaft 3 Assembly



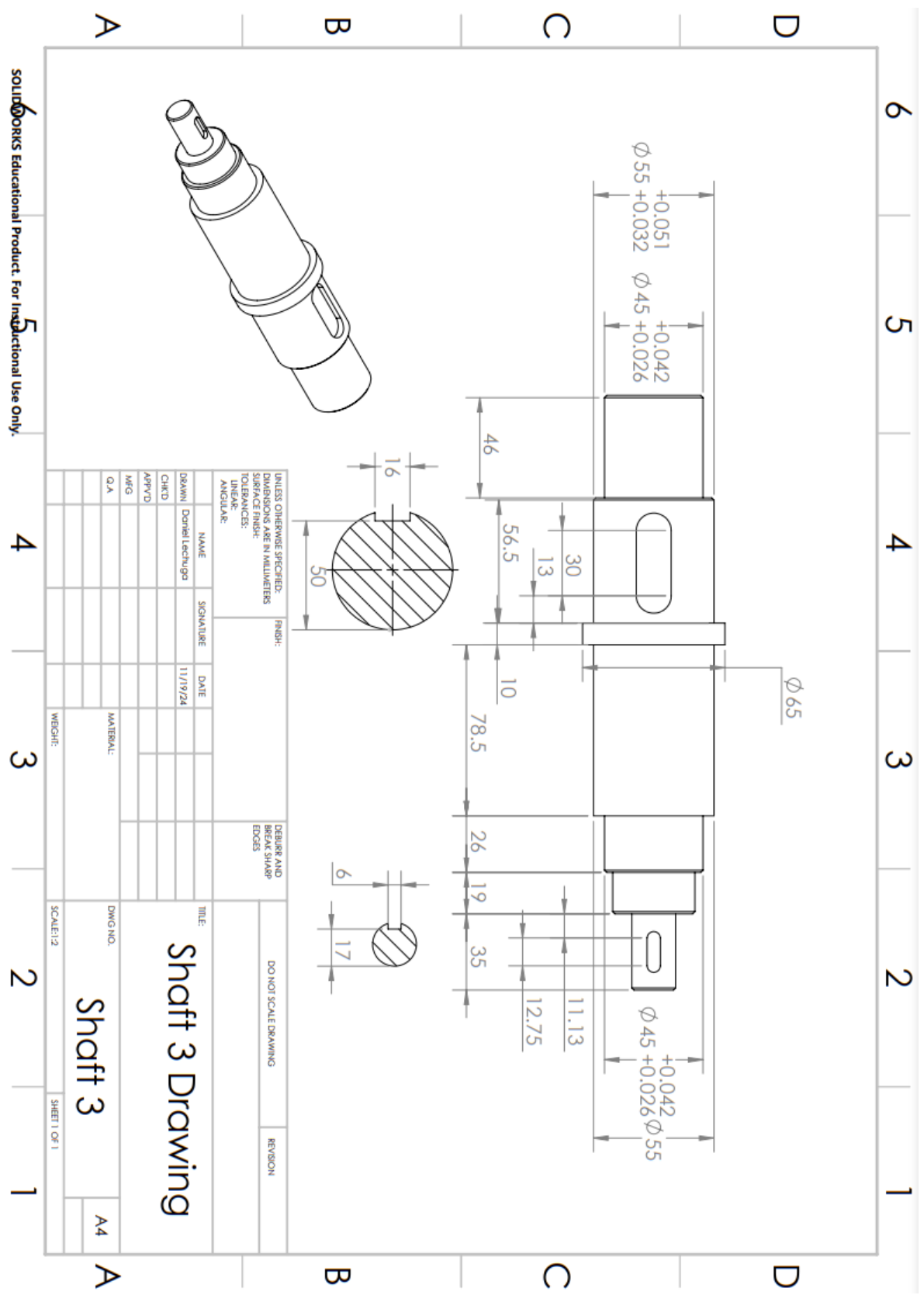
Shaft Part Drawings

Shaft 1



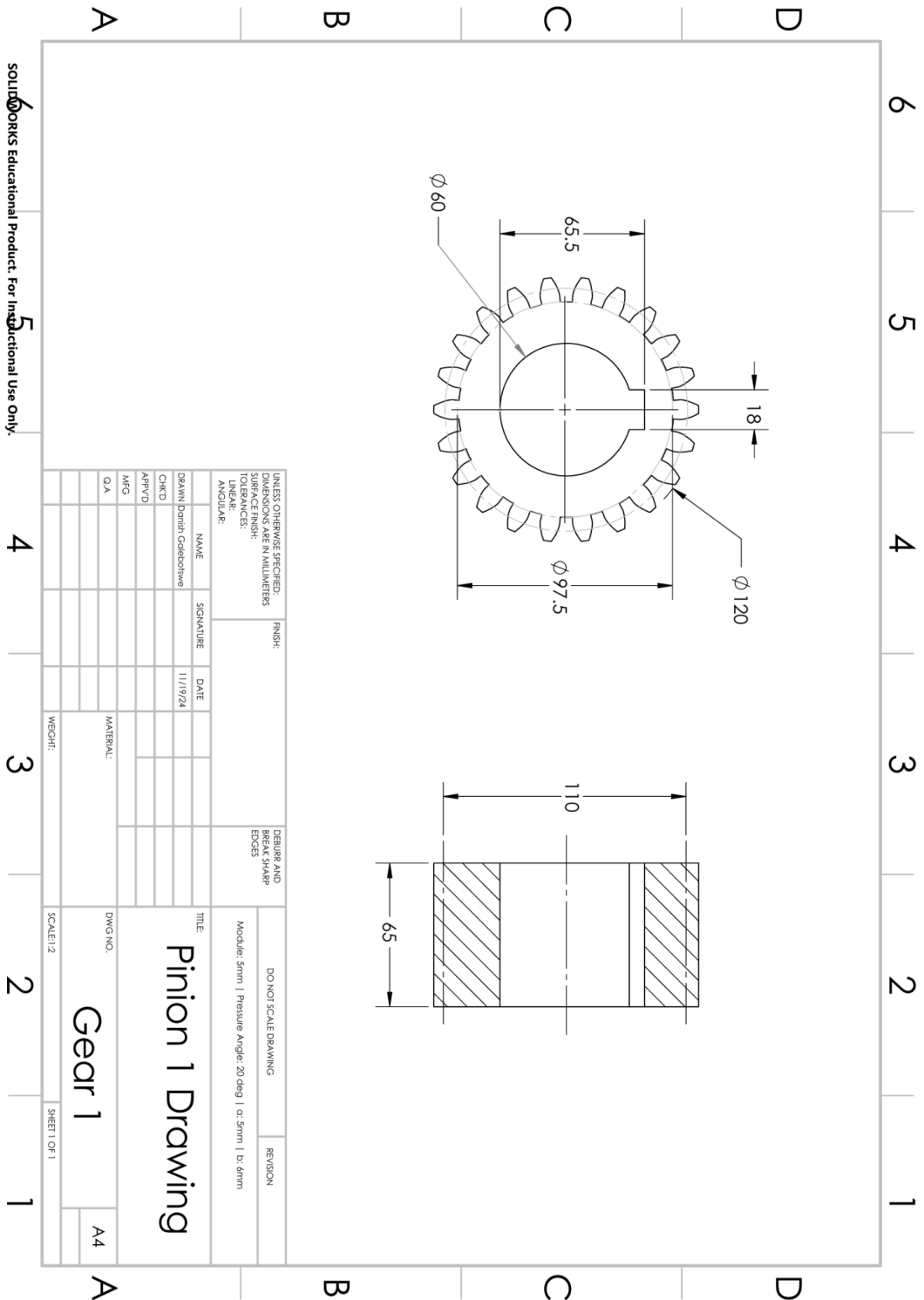


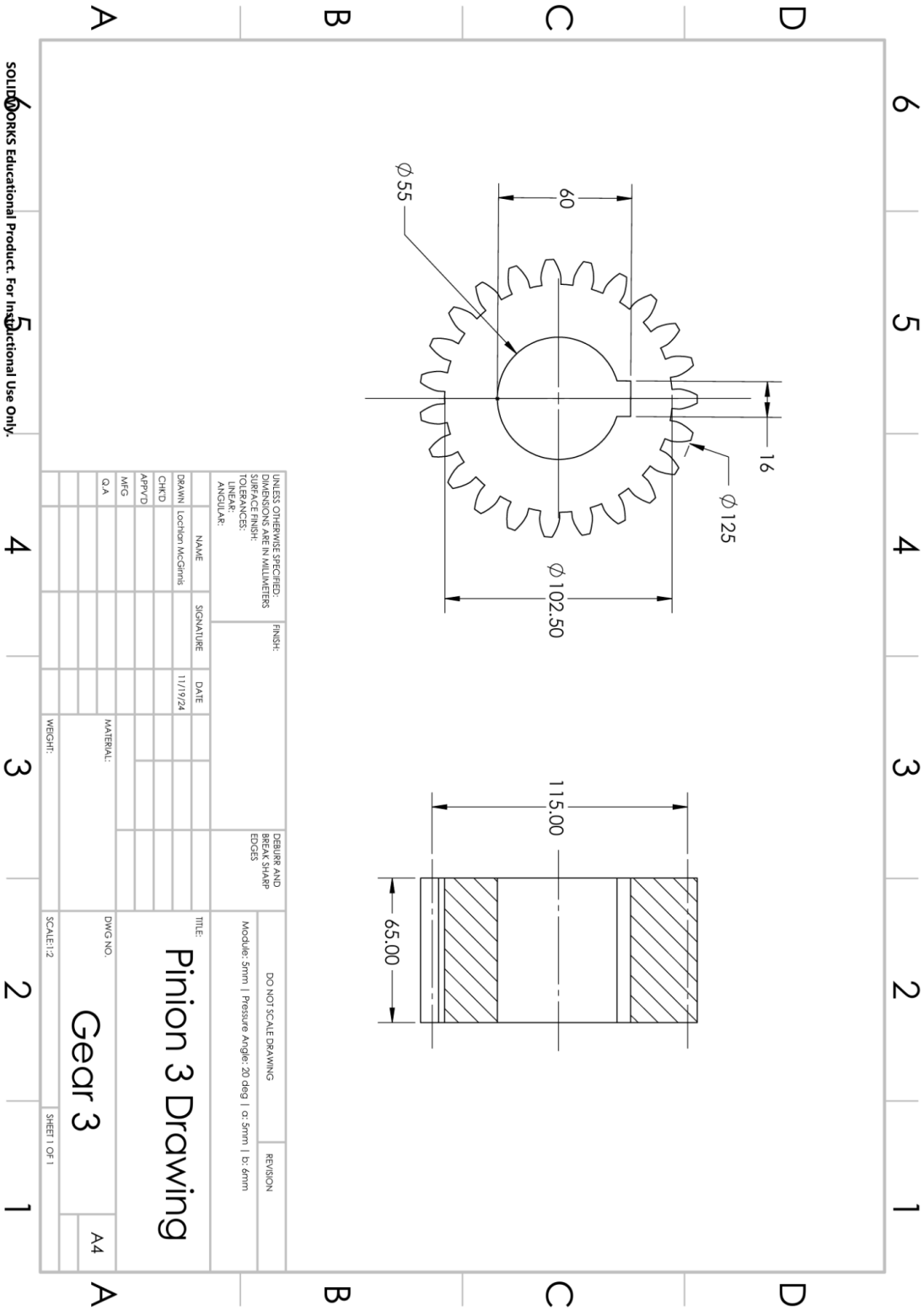
Shaft 3

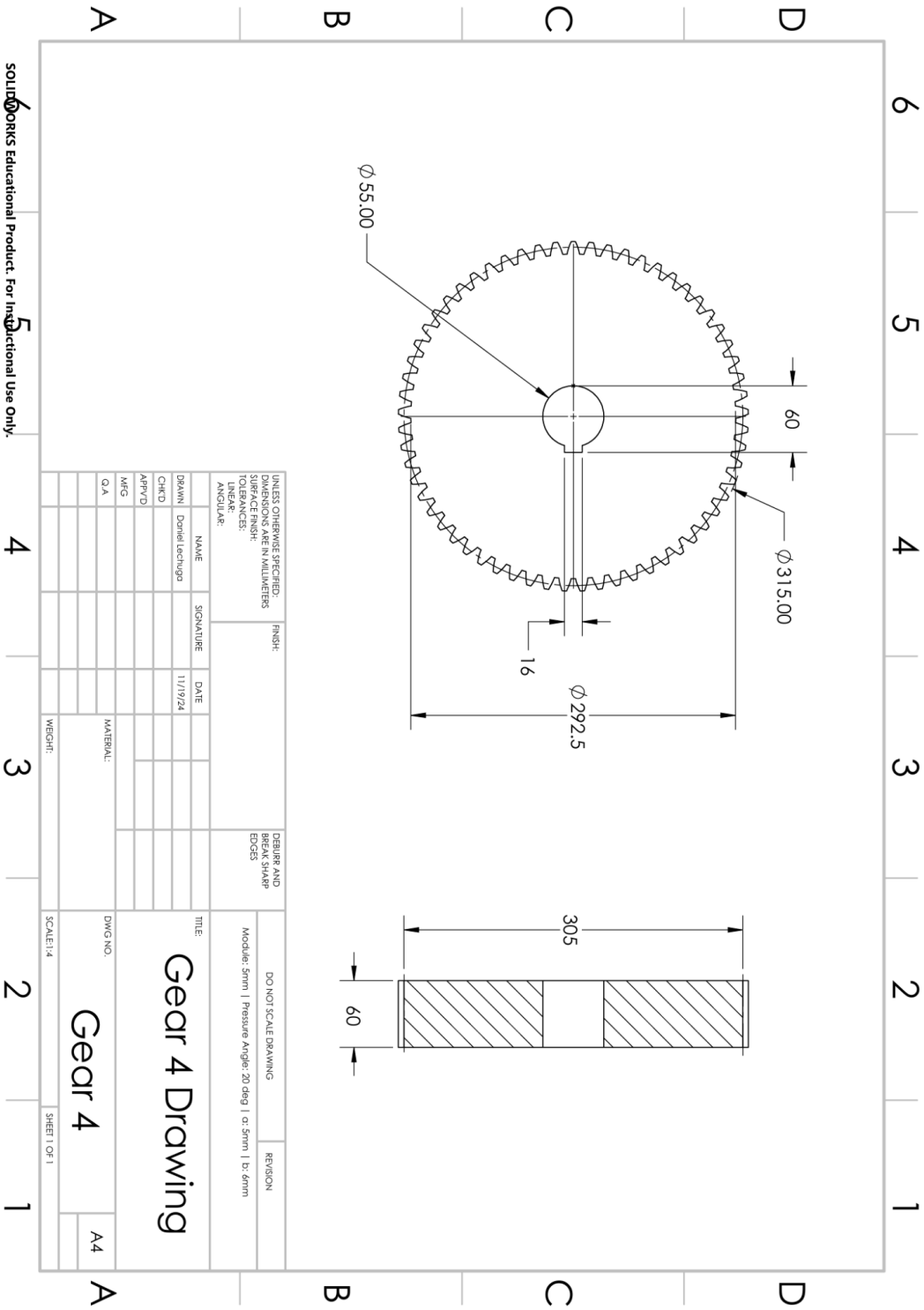


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Gear 1







Appendix 1 - Shaft Analysis

Shaft System 1:

Key Parameters

Parameter	Symbol	Value	Unit
Speed	n_l	2700	rpm
Input Power	H	60,000	W
Number of Teeth	N_l	22	-
Pitch Circle Diameter	d_{pl}	110	mm
Pressure Angle	ϕ	20	degrees

Equations and Calculations

1. Torque:

$$T_l = \frac{60H}{2\pi n_l} = \frac{60 \times 60,000}{2\pi \times 2700} = 212.21 \text{ Nm}$$

2. Tangential Force:

$$W_{tl} = \frac{2T_l}{d_{pl}} = \frac{212.21}{0.11} = 3858 \text{ N}$$

3. Radial Force:

$$W_{rl} = W_{tl} \tan(\phi) = 3858 \text{ N} \times \tan(20^\circ) = 1404 \text{ N}$$

4. Normal Force:

$$W_{nl} = \frac{W_t}{\cos(\phi)} = \frac{3858 \text{ N}}{\cos(20^\circ)} = 4106 \text{ N}$$

5. Velocity:

$$V_{l2} = \frac{\pi n_l d_{pl}}{60} = \frac{\pi \times 2700 \times 0.11}{60} = 15.6 \text{ m/s}$$

6. Operation Hours:

$$t = 5 \text{ years} \times 52 \text{ weeks/year} \times 7 \text{ days/week} \times 8 \text{ hours/day} = 14560 \text{ hours}$$

7. Number of Cycles:

$$N_{cl} = \frac{60 \cdot t \cdot n_l}{10^6} = \frac{60 \times 14560 \times 2700}{10^6} = 2358.72 \text{ Mc}$$

8. Input Gear Parameters:

$$\text{Pitch circle diameter: } d_{pi} = 780 \text{ mm}$$

$$\text{Tangential force: } W_i^t = \frac{T_l}{d_{pi}} = \frac{212.21 \text{ Nm}}{200 \text{ mm}} = 544 \text{ N}$$

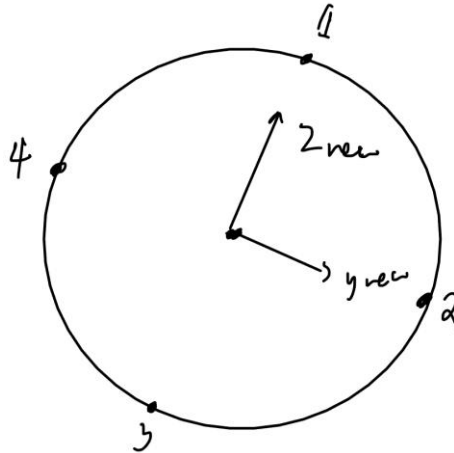
$$\text{Reaction Forces: } = W_i^t \tan(\phi) = 544 \text{ N} \times \tan(20 \text{ deg}) = 198 \text{ N}$$

Free Body Diagram and Bending Moments

- **Shaft Parameters:**

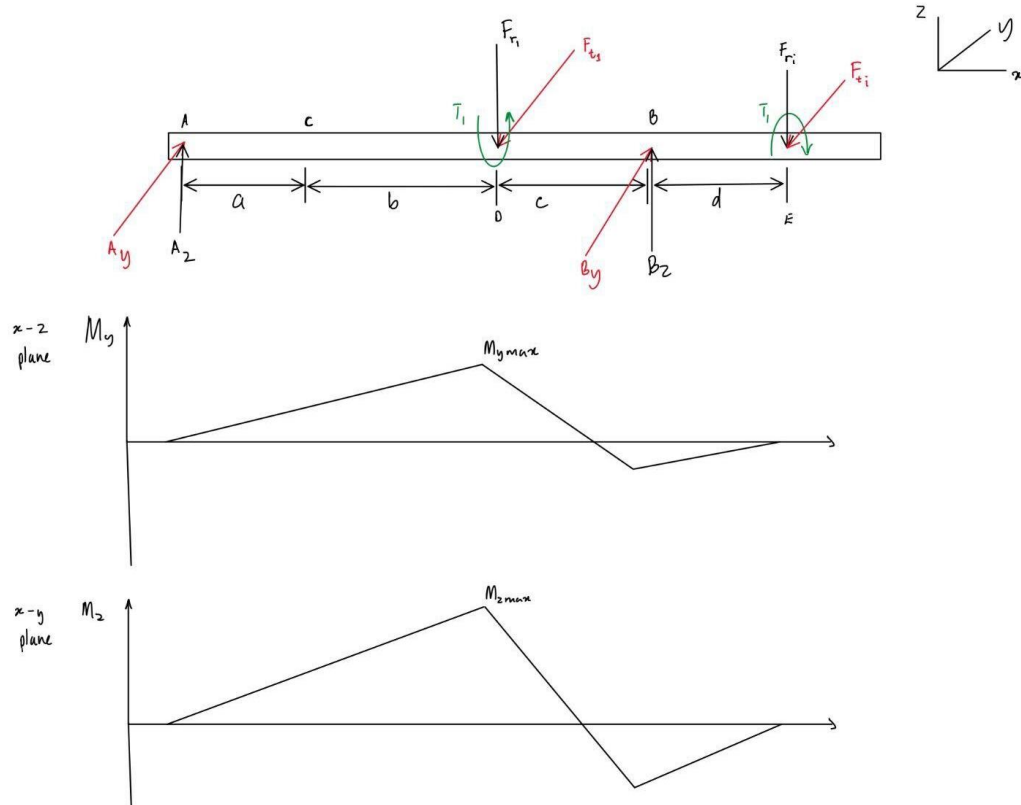
- $a = 58\text{mm}$
- $b = 72.5\text{mm}$
- $c = 55.5\text{mm}$
- $d = 20\text{mm}$

- Critical cross section: D



- Diameter at D: 50mm
- Critical Point: 3

- **Reaction Forces:**



- $\Sigma F_y = 0: A_y + B_y - W_t - W_i = 0$
- $\Sigma F_z = 0: A_z + B_z - W_r - W_i = 0$
- **Moments:**
 - $\Sigma M_y = 0(@A): B_z(a + b + c) - W_r(a + b) - W_i(a + b + c + d) = 0$
 - $\Sigma M_z = 0(@A): B_y(a + b + c) - W_t(a + b) - W_i(a + b + c + d) = 0$

● **Maximum Moments & Shear:**

$$M_{ymax} = A_z(a + b) = 52 \text{ Nm}$$

$$M_{zmax} = A_y(a + b) = 144 \text{ Nm}$$

$$M_{D-} = \sqrt{M_{ymax}^2 + M_{zmax}^2} = 153 \text{ Nm}$$

$$V_{D-} = \sqrt{A_y^2 + A_z^2} = 1171 \text{ N}$$

Stress Analysis

1. **Transverse Shear Stress:**

$$\tau_V = \frac{16V_{max}}{3\pi d_{cs}^2} = 0.8 \text{ MPa}$$

2. **Torsional Shear Stress:**

$$\tau_T = \frac{16T_l}{\pi d_{cs}^3} = 8.6 \text{ MPa}$$

3. **Bending Stress:**

$$\sigma_b = \frac{32M_{max}}{\pi d_{cs}^3} = 12.4 \text{ MPa}$$

4. **Von Mises Stresses:** Where $K_f = 1.5$, and $K_{fs} = 1.3$

○ **Alternating Von Mises Stress (σ_{vm-a}):**

$$\sigma_{vm-a} = \sqrt{(K_f \cdot \sigma_b)^2 + 3(K_{fs} \cdot \tau_V)^2} = \sqrt{(1.5 \cdot 12.4)^2 + 3(1.3 \cdot 0.8)^2} = 18.8 \text{ MPa}$$

○ **Mean Von Mises Stress (σ_{vm-m}):**

$$\sigma_{vm-m} = \sqrt{3 \cdot \tau_T^2} = \sqrt{3 \cdot 8.6^2} = 15.0 \text{ MPa}$$

○ **Peak Von Mises Stress ($\sigma_{vm-peak}$):**

$$\sigma_{vm-peak} = \sqrt{\sigma_b^2 + 3(\tau_T + \tau_V)^2} = \sqrt{12.4^2 + 3(8.6 + 0.8)^2} = 20.6 \text{ MPa}$$

Material Properties

The shaft material selected is **AISI 1020 steel**, with the following derived properties based on the design conditions:

1. **Ultimate Tensile Strength (S_{ut}):**

$$S_{ut} = 450 \text{ MPa}$$

2. **Yield Strength (S_y):**

$$S_y = 330 \text{ MPa}$$

3. **Endurance Limit (S_e):** The true endurance limit is calculated using modifying factors:

$$S_e = k_f k_s k_r k_t k_m S_e'$$

Where:

- k_f = Stress concentration factor = $1.58(S_{ut}/1000)^{-0.085} = 0.94$
- k_s = Size factor = $1.189(1000d_{cs})^{-0.112} = 0.76$
- k_r = Reliability factor = 0.82 (for 99% reliability)
- k_t = Temperature factor = 1 (ambient temperature)
- k_m = Miscellaneous factor = 1 (general use)

Substituting values:

$$S_e = 0.94 \times 0.80 \times 0.82 \times 1 \times 1 \times 450 = 133 \text{ MPa}$$

Factors of Safety

1. **Goodman Factor of Safety (n_s):** The Goodman Factor of Safety is given by:

$$n_s = \left(\frac{\sigma_{vm-a}}{S_e} + \frac{\sigma_{vm-m}}{S_{ut}} \right)^{-1} = \left(\frac{18.8}{133} + \frac{15.0}{450} \right)^{-1} = 5.74$$

2. **Yield Factor of Safety (n_y):** The Yield Factor of Safety is given by:

$$n_y = \frac{S_y}{\sigma_{vm-peak}} = \frac{330}{55} = 16.10$$

Shaft System 2:

Key Parameters

Parameter	Symbol	Value	Unit
Speed	$n_2 = n_3$	1060.71	rpm
Input Power	H	60,000	W
Shaft 2 Torque	T_2	540.16	Nm
Number of Teeth	N_2	56	-
Number of Teeth	N_3	23	-
Pitch Circle Diameter	d_{p2}	280	mm
Pitch Circle Diameter	d_{p3}	115	mm
Pressure Angle	ϕ	20	degrees

Equations and Calculations

9. Torque:

$$T_2 = T_3 = \frac{60H}{2\pi n_2} = \frac{60 \times 60,000}{2\pi \times 1060.71} = 540.16 \text{ Nm}$$

10. Tangential Force (Gear 2):

$$W_{t2} = \frac{2T_2}{d_{p2}} = \frac{2 \times 540.16}{0.280} = 3858 \text{ N}$$

11. Tangential Force (Gear 3):

$$W_{t3} = \frac{2T_3}{d_{p3}} = \frac{2 \times 540.16}{0.115} = 9394 \text{ N}$$

12. Radial Force (Gear 2):

$$W_{r2} = W_{t2} \tan(\phi) = 3858 \text{ N} * \tan(20^\circ) = 1404 \text{ N}$$

13. Radial Force (Gear 3):

$$W_{r3} = W_{t3} \tan(\phi) = 9394 \text{ N} * \tan(20^\circ) = 3419 \text{ N}$$

14. Normal Force (Gear 2):

$$W_n = \frac{W_t}{\cos(\phi)} = \frac{3858 \text{ N}}{\cos(20^\circ)} = 4106 \text{ N}$$

15. Normal Force (Gear 3):

$$W_n = \frac{W_t}{\cos(\phi)} = \frac{9394 \text{ N}}{\cos(20^\circ)} = 9997 \text{ N}$$

16. Velocity (Gear 2):

$$V_{l2} = \frac{\pi n_2 d_{p2}}{60} = \frac{\pi \times 1060.71 \times 0.280}{60} = 15.6 \text{ m/s}$$

17. Velocity (Gear 3):

$$V_{34} = \frac{\pi n_3 d_{p3}}{60} = \frac{\pi \times 1060.71 \times 0.115}{60} = 6.39 \text{ m/s}$$

18. Operation Hours:

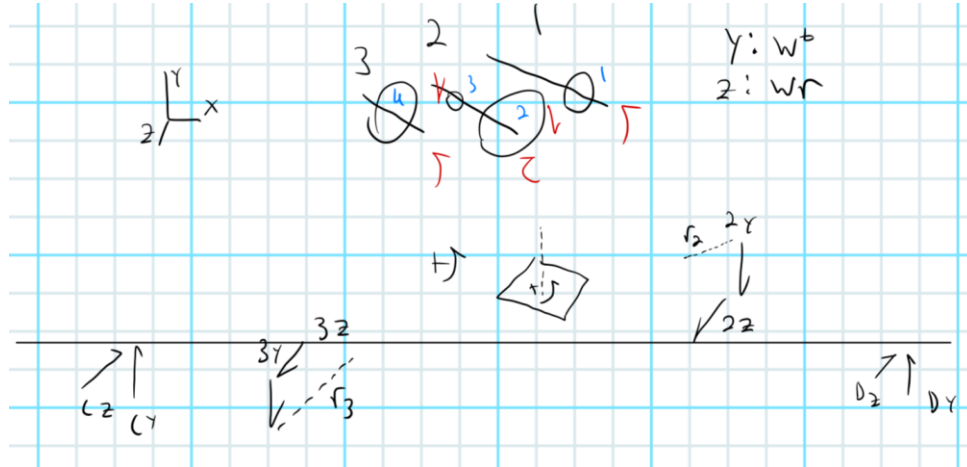
$$t = 5 \text{ years} \times 52 \text{ weeks/year} \times 7 \text{ days/week} \times 8 \text{ hours/day} = 14560 \text{ hours}$$

19. Number of Cycles:

$$N_{c2} = N_{c3} = \frac{60 \cdot t \cdot n_2}{10^6} = \frac{60 \times 14560 \times 1060.71}{10^6} = 926.64 \text{ Mc}$$

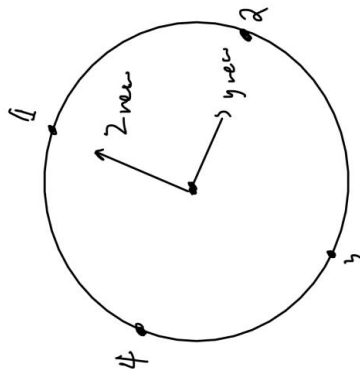
Free Body Diagram and Bending Moments

• **Diagram:**



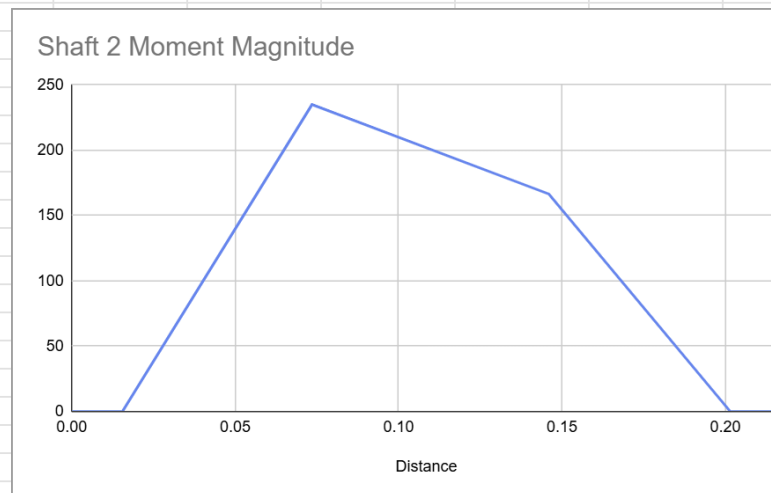
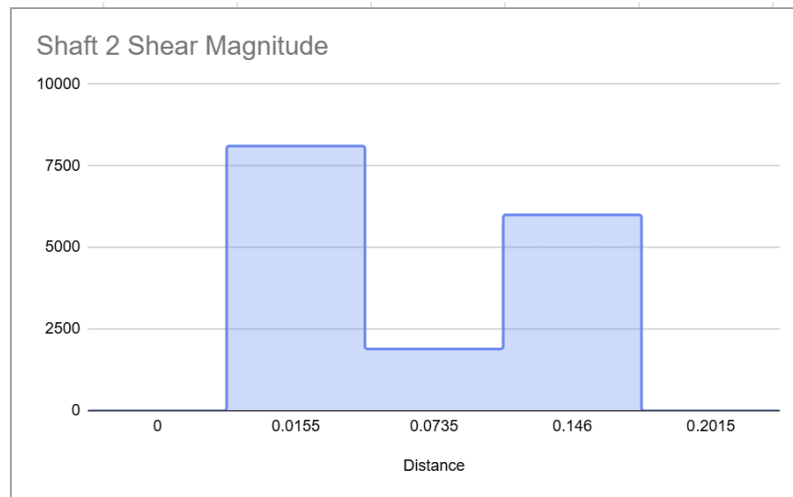
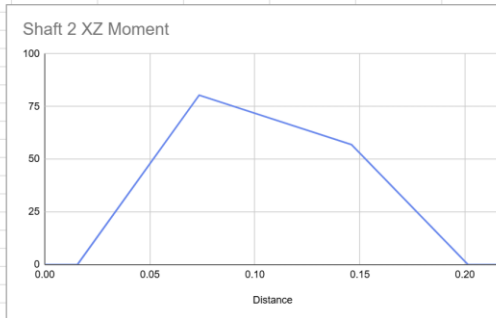
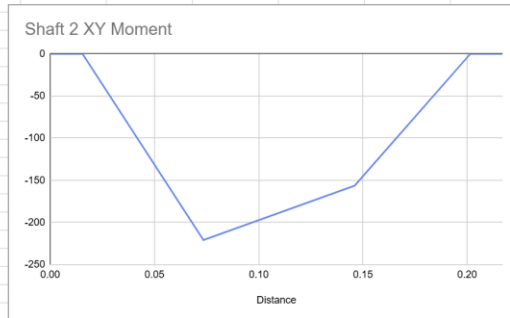
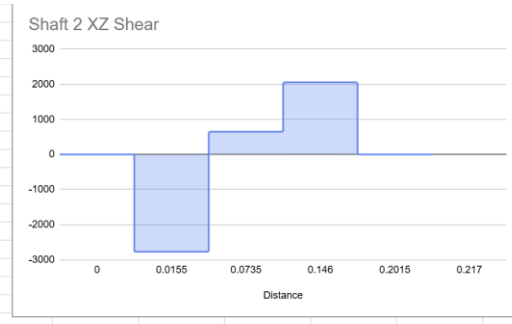
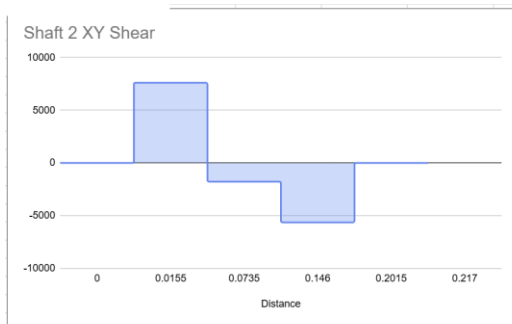
• **Shaft Parameters:**

- $a = 58 \text{ mm}$
- $b = 72.5 \text{ mm}$
- $c = 55.5 \text{ mm}$
- Critical cross section: Gear 3⁺ (Side closer to shaft center on force application point of gear 3 due to the presence of torque)



- Diameter at Gear 3⁺: $55 \text{ mm} = d_{cs}$
- Critical Point: 4

• **Reaction Forces:**



○ $\Sigma F_y = 0: C_y + D_y - W_{t1} - W_{t2} = 0$

- $\Sigma F_z = 0: C_z + D_z - W_{r1} - W_{r2} = 0$
- **Moments:**
 - $\Sigma M_y = 0(@C): -W_{r2}(a) - W_{r3}(a + b) + D_y(a + b + c) = 0$
 - $\Sigma M_z = 0(@C): -W_{t2}(a) - W_{t3}(a + b) + D_z(a + b + c) = 0$
- **Maximum Moments & Shear:**

$$M_{ymax} = A_y(a) = 221 \text{ Nm}$$

$$M_{zmax} = A_z(a) = 80.4 \text{ Nm}$$

$$M_{3+} = \sqrt{M_{ymax}^2 + M_{zmax}^2} = 235 \text{ Nm}$$

$$V_{3+} = \sqrt{A_y^2 + A_z^2} = 8105 \text{ N}$$

Stress Analysis

5. Torsional Shear Stress:

$$\tau_T = \frac{16T_2}{\pi d_{cs}^3} = 16.5 \text{ MPa}$$

6. Bending Stress:

$$\sigma_b = \frac{32M_{max}}{\pi d_{cs}^3} = 14.4 \text{ MPa}$$

7. Von Mises Stresses: Where $K_f = 1.5$, and $K_{fs} = 1.3$

- **Alternating Von Mises Stress (σ_{vm-a}):**

$$\sigma_{vm-a} = \sqrt{(K_f \cdot \sigma_b)^2 + 3(K_{fs} \cdot \tau_T)^2} = \sqrt{(1.5 \cdot 14.4)^2 + 3(1.3 \cdot 0)^2} = 21.6 \text{ MPa}$$

- **Mean Von Mises Stress (σ_{vm-m}):**

$$\sigma_{vm-m} = \sqrt{3 \cdot \tau_T^2} = \sqrt{3 \cdot 16.5^2} = 28.6 \text{ MPa}$$

- **Peak Von Mises Stress ($\sigma_{vm-peak}$):**

$$\sigma_{vm-peak} = \sqrt{(\sigma_b + \sigma_m)^2 + 3(\tau_T + \tau_V)^2} = \sqrt{14.4^2 + 3(16.5)^2} = 32 \text{ MPa}$$

Material Properties

The shaft material selected is **AISI 1020 steel**, with the following derived properties based on the design conditions:

4. Ultimate Tensile Strength (S_{ut}):

$$S_{ut} = 450 \text{ MPa}$$

5. Yield Strength (S_y):

$$S_y = 330 \text{ MPa}$$

6. Endurance Limit (S_e): The true endurance limit is calculated using modifying factors:

$$S_e = k_f k_s k_r k_t k_m S'_e$$

Where:

- k_f = Stress concentration factor = $1.58(S_{ut}/1000)^{-0.085} = 0.94$
- k_s = Size factor = $1.189(1000d_{cs})^{-0.112} = 0.76$
- k_r = Reliability factor = 0.82 (for 99% reliability)
- k_t = Temperature factor = 1 (ambient temperature)
- k_m = Miscellaneous factor = 1 (general use)

Substituting values:

$$S_e = 0.94 \times 0.80 \times 0.82 \times 1 \times 1 \times (450 \times 0.5) = 132 \text{ MPa}$$

Factors of Safety

3. **Goodman Factor of Safety (n_s):** The Goodman Factor of Safety is given by:

$$n_s = \left(\frac{\sigma_{vm-a}}{S_e} + \frac{\sigma_{vm-m}}{S_{ut}} \right)^{-1} = \left(\frac{21.6}{132} + \frac{28.6}{450} \right)^{-1} = 4.39$$

4. **Yield Factor of Safety (n_y):** The Yield Factor of Safety is given by the following (ignoring stress concentrations):

$$n_y = \left(\frac{\sigma_{vm-a}}{S_y} + \frac{\sigma_{vm-m}}{S_y} \right)^{-1} = \left(\frac{14.4}{330} + \frac{28.6}{330} \right)^{-1} = 7.70$$

Shaft System 3:

Key Parameters

Parameter	Symbol	Value	Unit
Speed	n_4	399.94	rpm
Module	m	5	mm
Shaft 1 Torque	T_1	212.21	Nm
Train value	i	6.75	-
Number of Teeth	N_4	61	-
Pitch Circle Diameter	d_{p4}	305	mm
Pressure Angle	ϕ	20	degrees

Equations and Calculations

20. Torque:

$$T_4 = T_1 \cdot i = 212.21 \cdot 6.75 = 1432.60 \text{ Nm}$$

21. Tangential Force:

$$W_t = \frac{2T_4}{d_{p4}} = \frac{2 \cdot 1432.60}{0.305} = 9394.13 \text{ N}$$

22. Radial Force:

$$W_r = W_t \tan(\phi) = 9394.13 \text{ N} \cdot \tan(20^\circ) = 3419.18 \text{ N}$$

23. Normal Force:

$$W_n = \frac{W_t}{\cos(\phi)} = \frac{9394.13 \text{ N}}{\cos(20^\circ)} = 4106 \text{ N}$$

24. Velocity:

$$V_{34} = \frac{\pi n_4 d_{p4}}{60} = \frac{\pi \times 399.94 \times 0.305}{60} = 6.39 \text{ m/s}$$

25. Operation Hours:

$$t = 5 \text{ years} \times 52 \text{ weeks/year} \times 7 \text{ days/week} \times 8 \text{ hours/day} = 14560 \text{ hours}$$

26. Number of Cycles:

$$N_{c4} = \frac{60 \cdot t \cdot n_4}{10^6} = \frac{60 \times 14560 \times 399.94}{10^6} = 349.39 \text{ Mc}$$

27. Input Gear Parameters:

$$\text{Pitch circle diameter: } d_{pi} = 780 \text{ mm}$$

$$\text{Tangential force: } W_j^t = \frac{2T_4}{d_{pi}} = \frac{2 \cdot 1432.60 \text{ Nm}}{610 \text{ mm}} = 3673.34 \text{ N}$$

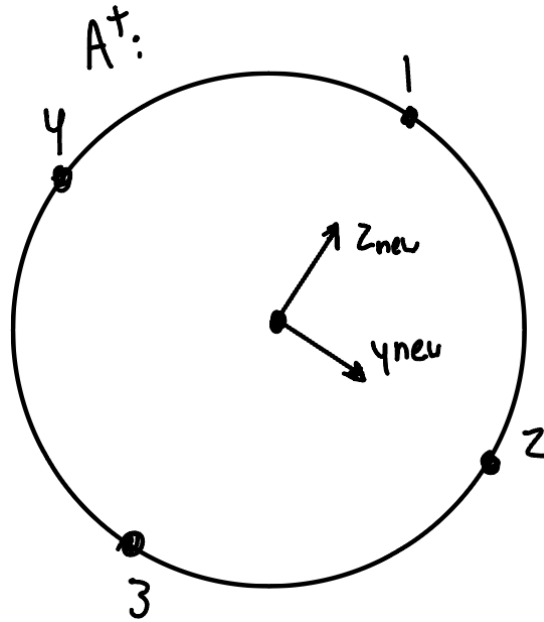
$$\text{Reaction Forces: } W_j^r = W_j^t \tan(\phi) = 3673.34 \text{ N} \times \tan(20 \text{ deg}) = 1336.99 \text{ N}$$

Free Body Diagram and Bending Moments

- **Shaft Parameters:**

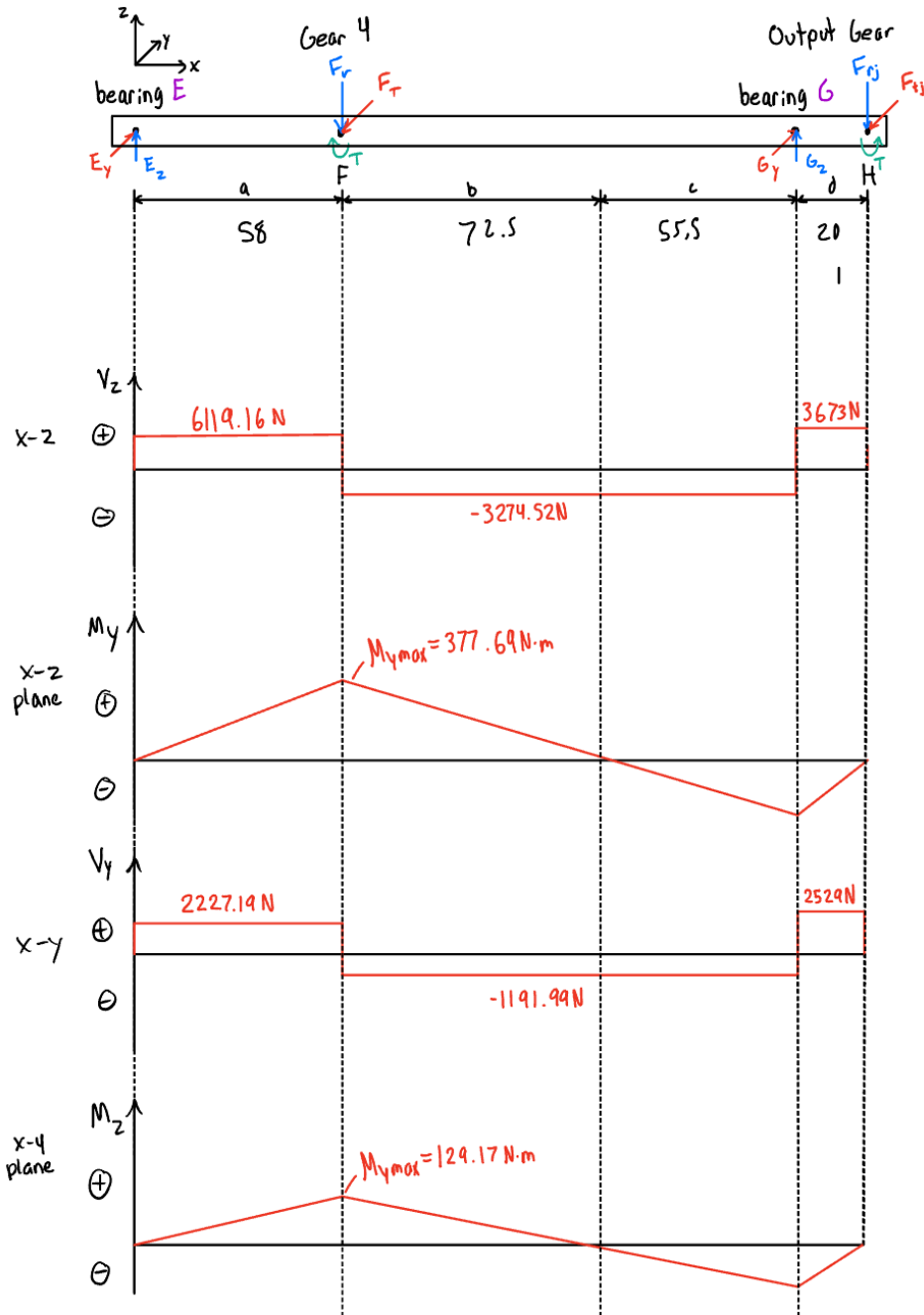
- $a = 58\text{mm}$
- $b = 72.5\text{mm}$
- $c = 55.5\text{mm}$
- $d = 20\text{mm}$

- Critical cross section: D^-



- Diameter at F^+ : 55mm
- Critical Point: 3

- **Reaction Forces:**



- $\Sigma F_y = 0: E_y + G_y - W_t - W_{tj} = 0$

- $\Sigma F_z = 0: A_z + B_z - W_r - W_{rj} = 0$

- Moments:**

- $\Sigma M_y = 0(@E): G_z(a + b + c) - W_r(a) - W_{rj}(a + b + c + d) = 0$

- $\Sigma M_z = 0(@E): G_y(a + b + c) - W_t(a) - W_{tj}(a + b + c + d) = 0$

- Maximum Moments & Shear:**

$$M_{y\max} = E_z(a) = 354.91 \text{ Nm}$$

$$M_{z\max} = E_y(a) = 129.18 \text{ Nm}$$

$$M_{max} = M_{F+} = \sqrt{M_{y_{max}}^2 + M_{z_{max}}^2} = 377.69 Nm$$

$$V_{max} = V_{F+} = \sqrt{E_y^2 + E_z^2} = 6511.88 N$$

Stress Analysis

1. **Transverse Shear Stress:**

$$\tau_V = \frac{16V_{max}}{3\pi d_{F+}^2} = 3.65 MPa$$

2. **Torsional Shear Stress:**

$$\tau_T = \frac{16T_l}{\pi d_{F+}^3} = 43.85 MPa$$

3. **Bending Stress:**

$$\sigma_b = \frac{32M_{max}}{\pi d_{F+}^3} = 23.12 MPa$$

4. **Von Mises Stresses:** Where $K_f = 1.5$, and $K_{fs} = 1.3$

- **Alternating Von Mises Stress (σ_{vm-a}):**

$$\sigma_{vm-a} = \sqrt{(K_f \cdot \sigma_b)^2 + 3(K_{fs} \cdot \tau_V)^2} = \sqrt{(1.5 \cdot 23.12)^2 + 3(1.3 \cdot 3.65)^2} = 35.65 MPa$$

- **Mean Von Mises Stress (σ_{vm-m}):**

$$\sigma_{vm-m} = \sqrt{3 \cdot \tau_T^2} = \sqrt{3 \cdot 43.85^2} = 75.96 MPa$$

- **Peak Von Mises Stress ($\sigma_{vm-peak}$):**

$$\sigma_{vm-peak} = \sqrt{\sigma_b^2 + 3(\tau_T + \tau_V)^2} = \sqrt{23.12^2 + 3(43.85 + 3.65)^2} = 85.47 MPa$$

Material Properties

The shaft material selected is **AISI 1020 steel**, with the following derived properties based on the design conditions:

7. **Ultimate Tensile Strength (S_{ut}):**

$$S_{ut} = 450 MPa$$

8. **Yield Strength (S_y):**

$$S_y = 330 MPa$$

9. **Endurance Limit (S_e):** The true endurance limit is calculated using modifying factors:

$$S_e = k_f k_s k_r k_t k_m S_e'$$

Where:

- k_f = Stress concentration factor = $1.58(S_{ut}/1000)^{-0.085} = 0.94$
- k_s = Size factor = $1.189(1000d_{cs})^{-0.112} = 0.76$
- k_r = Reliability factor = 0.82 (for 99% reliability)
- k_t = Temperature factor = 1 (ambient temperature)
- k_m = Miscellaneous factor = 1 (general use)

Substituting values:

$$S_e = 0.94 \times 0.76 \times 0.82 \times 1 \times 1 \times 450 = 132 MPa$$

Factors of Safety

5. **Goodman Factor of Safety (n_s):** The Goodman Factor of Safety is given by:

$$n_s = \left(\frac{\sigma_{vm-a}}{S_e} + \frac{\sigma_{vm-m}}{S_{ut}} \right)^{-1} = \left(\frac{35.65}{132} + \frac{75.96}{450} \right)^{-1} = 2.27$$

6. **Yield Factor of Safety (n_y):** The Yield Factor of Safety is given by:

$$n_y = \frac{S_y}{\sigma_{vm-peak}} = \frac{330}{85.47} = 3.86$$

Appendix 2 - Gear Analysis

Shaft 1 Gear Analysis:

Key Parameters

Parameter	Symbol	Equation	Value	Unit
Train Value	i	$i = 6.75$	6.75	-
Input Power	H	$H = 60,000$	60,000	W
Torque	T_l	$T_l = \frac{60H}{2\pi n_l}$	212.21	Nm
Number of Teeth (Gear 1)	N_l	$N_l = 22$	22	-
Addendum	a	$a = m$	5	mm
Dedendum	b	$b = 1.25m$	6	mm
Tooth Height	h	$h = a + b$	11	mm
Clearance	c	$c = b - a$	1	mm
Circular Pitch	p_c	$p_c = \pi m$	16	mm
Base Pitch	p_b	$p_b = \pi m \cos(\phi)$	15	mm
Base Width	b_w	$b_w = 13m$	65	mm
Tooth Thickness	t	$t = \frac{p_c}{2}$	8	mm
Pitch Circle Diameter	d_{pl}	$d_{pl} = N_l \cdot m$	110	mm
Outside Circle Diameter	d_{ol}	$d_{ol} = d_{pl} + 2a$	120	mm
Base Circle Diameter	d_{bl}	$d_{bl} = d_{pl} \cos(\phi)$	103	mm
Root Circle Diameter	d_{rl}	$d_{rl} = d_{pl} - 2b$	97.5	mm
Tangential Force	W_t	$W_t = \frac{T_l}{d_{pl}/2}$	3858	N
Velocity	V_{l2}	$V_{l2} = \frac{\pi n_l d_{pl}}{60}$	15.6	m/s
Operation Hours	t	$t = 5 \text{ years} \times 52 \times 7 \times 8$	14,560	hrs

Number of Cycles	N_{cl}	$N_{cl} = \frac{60tn_l}{10^6}$	2358.72	Mc
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Equations and Calculations

1. **Radial Force (W_r):** The radial force is calculated using the pressure angle:

$$W_r = W_t \cdot \tan(\phi)$$

Substituting values:

$$W_r = 3858 \cdot \tan(20^\circ) = 1404 \text{ N}$$

2. **Normal Force (W_n):** The normal force is derived from the tangential force:

$$W_n = \frac{W_t}{\cos(\phi)}$$

Substituting values:

$$W_n = \frac{3858}{\cos(20^\circ)} = 4106 \text{ N}$$

Stress Analysis

1. **Bending Stress (σ_{pb}):** The bending stress is calculated using:

$$\sigma_{pb} = \frac{W_t \cdot k_b}{b_w \cdot m \cdot Y_j}$$

Where:

- o Bending geometry factor: $Y_j = 0.33$ (from Figure 5-50 in chapter 5 e-book)
- o Load factor:

$$k_b = K_b \cdot K_i \cdot K_a \cdot K_m \cdot K_v$$

$$\bullet \quad K_b = 1, K_i = 1, K_a = 1, K_m = 1.2, K_v = \frac{59.7 + \sqrt{200V_{12}}}{59.7}^{0.825}$$

Substituting Velocity $V_{12} = 15.6 \text{ m/s}$:

$$K_v = \left(\frac{59.7 + \sqrt{200 \cdot 15.6}}{59.7} \right)^{0.825} = 1.72$$

$$k_b = 1 \cdot 1 \cdot 1 \cdot 1.2 \cdot 1.72 = 2.07$$

Substituting into the stress equation:

$$\sigma_{pb} = \frac{3858 \cdot 2.07}{65 \cdot 5 \cdot 0.33} = 74 \text{ MPa}$$

2. **Contact Stress (σ_{pc}):** The contact stress is given by:

$$\sigma_{pc} = K_e \cdot \sqrt{\frac{W_t \cdot k_c}{b_w \cdot d_{pl} \cdot I}}$$

Where:

$$K_e = \sqrt{\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2}}$$

With material properties:

$$\circ \quad E_1 = E_2 = 2.07 \times 10^{11} \text{ Pa}, \nu_1 = \nu_2 = 0.3$$

$$K_e = \sqrt{\frac{1 - 0.3^2}{2.07 \times 10^{11}} + \frac{1 - 0.3^2}{2.07 \times 10^{11}}} = 473675$$

Geometry factor:

$$I = \pi \cdot \cos(\phi) \cdot \sin(\phi) \left(1 + \frac{d_{p1}}{d_{p2}} \right)$$

Substituting:

$$I = \pi \cdot \cos(20^\circ) \cdot \sin(20^\circ) \cdot \left(1 + \frac{110}{280} \right) = 0.72$$

Load factor:

$$k_c = 2.07$$

Substituting into the contact stress equation:

$$\sigma_{pc} = 473675 \cdot \sqrt{\frac{3858 \cdot 2.07}{65 \cdot 110 \cdot 0.72}} = 587 \text{ MPa}$$

Material Properties and Allowable Stresses

1. **Hardness:**

$$HB_p = 578$$

2. **Allowable Bending Stress (No Factors):**

$$S_{pb} = (0.703 \cdot HB_p + 113) \cdot 10^6 = 519 \text{ MPa}$$

3. **Allowable Contact Stress (No Factors):**

$$S_{pc} = (2.41 \cdot HB_p + 237) \cdot 10^6 = 1629 \text{ MPa}$$

4. **Geometry Factor for Bending:** Equation determined from Figure 5-46 in e-book.

$$Y_{np} = 1.6831 \cdot (N_{cl}^{-0.0323}) = 0.84$$

5. **Geometry Factor for Contact:** Equation determined from Figure 5-46 in e-book.

$$Z_{np} = 2.466 \cdot (N_{cl}^{-0.056}) = 0.74$$

6. **Load Sharing Factor:**

$$C_{hg} = 1 + ((8.89 \cdot 10^{-3}) \cdot \frac{HB_p}{HG_G} - 8.29 \cdot 10^{-3}) \left(\frac{N_G}{N_p} - 1 \right) = 1.005$$

7. **Allowable Bending Stress (With Factors):**

$$S_{all,bp} = \frac{S_{pb} \cdot Y_{np}}{K_t \cdot K_r} = \frac{519 \cdot 0.84}{1 \cdot 1} = 435 \text{ MPa}$$

8. **Allowable Contact Stress (With Factors):**

$$S_{all,cp} = \frac{S_{pc} \cdot Z_{np} \cdot C_{hp}}{K_t \cdot K_r} = \frac{1629 \cdot 0.74 \cdot 1}{1 \cdot 1} = 1200 \text{ MPa}$$

Factors of Safety

- **Bending Stress Factor of Safety (n_{bp}):**

$$n_{bp} = \frac{S_{all,bp}}{\sigma_{pb}} = \frac{435}{74} = 5.85$$

- **Contact Stress Factor of Safety (n_{cp}):**

$$n_{cp} = \frac{S_{all, cp}}{\sigma_{pc}} \frac{1200}{587} = 2.04$$

Shaft 2 Gear Analysis:

Key Parameters

Parameter	Symbol	Equation	Value	Unit
Train Value	i	$i = 6.75$	6.75	-
Input Power	H	$H = 60,000$	60,000	W
Torque	T_2	$T_2 = \frac{60H}{2\pi n_2}$	540.16	Nm
Number of Teeth (Gear 2)	N_2	$N_2 = 56$	56	-
Number of Teeth (Gear 3)	N_3	$N_3 = 23$	23	-
Addendum	a	$a = m$	5	mm
Dedendum	b	$b = 1.25m$	6	mm
Tooth Height	h	$h = a + b$	11	mm
Clearance	c	$c = b - a$	1	mm
Circular Pitch	p_c	$p_c = \pi m$	16	mm
Base Pitch	p_b	$p_b = \pi m \cos(\phi)$	15	mm
Base Width (Gear 2)	b_w	$b_w = 12m$	60	mm
Base Width (Gear 3)	b_w	$b_w = 13m$	65	mm
Tooth Thickness	t	$t = \frac{p_c}{2}$	8	mm
Pitch Circle Diameter (Gear 2)	d_{p2}	$d_{p2} = N_2 \cdot m$	280	mm
Outside Circle Diameter (Gear 2)	d_{o2}	$d_{o2} = d_{p2} + 2a$	290	mm
Base Circle Diameter (Gear 2)	d_{b2}	$d_{b2} = d_{p2} \cos(\phi)$	263	mm
Root Circle Diameter (Gear 2)	d_{r2}	$d_{r2} = d_{p2} - 2b$	267.5	mm
Tangential Force (Gear 2)	W_t	$W_t = \frac{T_l}{d_{p2}/2}$	3858	N

Velocity (Gear 2)	V_{l2}	$V_{l2} = \frac{\pi n_2 d_{p2}}{60}$	15.6	m/s
Pitch Circle Diameter (Gear 3)	d_{p3}	$d_{p3} = N_3 \cdot m$	115	mm
Outside Circle Diameter (Gear 3)	d_{o3}	$d_{o3} = d_{p3} + 2a$	125	mm
Base Circle Diameter (Gear 3)	d_{b3}	$d_{b3} = d_{p3} \cos(\phi)$	108	mm
Root Circle Diameter (Gear 3)	d_{r3}	$d_{r3} = d_{p3} - 2b$	102.5	mm
Tangential Force (Gear 3)	W_t	$W_t = \frac{T_l}{d_{p3}/2}$	9394	N
Velocity (Gear 3)	V_{34}	$V_{34} = \frac{\pi n_3 d_{p3}}{60}$	6.39	m/s
Operation Hours	t	$t = 5 \text{ years} \times 52 \times 7 \times 8$	14,560	hrs
Number of Cycles	N_c	$N_{c2} = N_{c3} = \frac{60 t n_2}{10^6}$	926.64	Mc

Equations and Calculations

3. **Radial Force (W_r):** The radial force is calculated using the pressure angle:

$$W_r = W_t \cdot \tan(\phi)$$

Substituting values:

$$W_{r2} = 3858 \cdot \tan(20^\circ) = 1404 \text{ N}$$

$$W_{r3} = 9394 \cdot \tan(20^\circ) = 3419 \text{ N}$$

4. **Normal Force (W_n):** The normal force is derived from the tangential force:

$$W_n = \frac{W_t}{\cos(\phi)}$$

Substituting values:

$$W_{n2} = \frac{3858}{\cos(20^\circ)} = 4106 \text{ N}$$

$$W_{n3} = \frac{9394}{\cos(20^\circ)} = 9997 \text{ N}$$

Stress Analysis (Gear 2)

3. **Bending Stress (σ_{pb}):** The bending stress is calculated using:

$$\sigma_{pb} = \frac{W_t \cdot k_b}{b_w \cdot m \cdot Y_j}$$

Where:

- o Bending geometry factor: $Y_j = 0.41$ (from Figure 5-50 in chapter 5 e-book)
- o Load factor:

$$k_b = K_b \cdot K_i \cdot K_a \cdot K_m \cdot K_v$$

$$\blacksquare K_b = 1, K_i = 1, K_a = 1, K_m = 1.15, K_v = \frac{59.7 + (200V_{12})^{0.5}}{59.7}^{0.825}$$

Substituting Velocity $V_{12} = 15.6 \text{ m/s}$:

$$K_v = \left(\frac{59.7 + \sqrt{200 \cdot 15.6}}{59.7} \right)^{0.825} = 1.72$$

$$k_b = 1 \cdot 1 \cdot 1 \cdot 1.2 \cdot 1.72 = 1.98$$

Substituting into the stress equation:

$$\sigma_{pb} = \frac{3858 \cdot 1.98}{60E - 3 \cdot 5E - 3 \cdot 0.41} = 62 \text{ MPa}$$

4. **Contact Stress (σ_{pc}):** The contact stress is given by:

$$\sigma_{pc} = K_e \cdot \sqrt{\frac{W_t \cdot k_c}{b_w \cdot d_{p2} \cdot I}}$$

Where:

$$K_e = \sqrt{\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2}}$$

With material properties:

$$\circ E_1 = E_2 = 2.07 \times 10^{11} \text{ Pa}, \nu_1 = \nu_2 = 0.3$$

$$K_e = \sqrt{\frac{1 - 0.3^2}{2.07 \times 10^{11}} + \frac{1 - 0.3^2}{2.07 \times 10^{11}}} = 473675$$

Geometry factor:

$$I = \pi \cdot \cos(\phi) \cdot \sin(\phi) \left(1 + \frac{d_{p1}}{d_{p2}} \right)$$

Substituting:

$$I = \pi \cdot \cos(20^\circ) \cdot \sin(20^\circ) \cdot \left(1 + \frac{110}{280} \right) = 0.72$$

Load factor:

$$k_c = 1.98$$

Substituting into the contact stress equation:

$$\sigma_{pc} = 473675 \cdot \sqrt{\frac{3858 \cdot 1.98}{60E - 3 \cdot 110E - 3 \cdot 0.41}} = 599 \text{ MPa}$$

Stress Analysis (Gear 3)

5. **Bending Stress (σ_{pb}):** The bending stress is calculated using:

$$\sigma_{pb} = \frac{W_t \cdot k_b}{b_w \cdot m \cdot Y_j}$$

Where:

- o Bending geometry factor: $Y_j = 0.35$ (from Figure 5-50 in chapter 5 e-book)
- o Load factor:

$$k_b = K_b \cdot K_i \cdot K_a \cdot K_m \cdot K_v$$

$$\blacksquare K_b = 1, K_i = 1, K_a = 1, K_m = 1.15, K_v = \frac{59.7 + (200V_{34})^{0.5}}{59.7}$$

Substituting Velocity $V_{34} = 6.39 \text{ m/s}$:

$$K_v = \left(\frac{59.7 + \sqrt{200 \cdot 6.39}}{59.7} \right)^{0.825} = 1.47$$

$$k_b = 1 \cdot 1 \cdot 1 \cdot 1.15 \cdot 1.47 = 1.69$$

Substituting into the stress equation:

$$\sigma_{pb} = \frac{9394 \cdot 1.47}{65E - 3 \cdot 5E - 3 \cdot 0.35} = 140 \text{ MPa}$$

6. **Contact Stress (σ_{pc}):** The contact stress is given by:

$$\sigma_{pc} = K_e \cdot \sqrt{\frac{W_t \cdot k_c}{b_w \cdot d_{p3} \cdot I}}$$

Where:

$$K_e = \sqrt{\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2}}$$

With material properties:

$$\circ E_1 = E_2 = 2.07 \times 10^{11} \text{ Pa}, \nu_1 = \nu_2 = 0.3$$

$$K_e = \sqrt{\frac{1 - 0.3^2}{2.07 \times 10^{11}} + \frac{1 - 0.3^2}{2.07 \times 10^{11}}} = 473675$$

Geometry factor:

$$I = \pi \cdot \cos(\phi) \cdot \sin(\phi) \left(1 + \frac{d_{p3}}{d_{p4}} \right)$$

Substituting:

$$I = \pi \cdot \cos(20^\circ) \cdot \sin(20^\circ) \cdot \left(1 + \frac{115}{305} \right) = 0.73$$

Load factor:

$$k_c = 1.69$$

Substituting into the contact stress equation:

$$\sigma_{pc} = 473675 \cdot \sqrt{\frac{9394 \cdot 1.69}{65E - 3 \cdot 115E - 3 \cdot 0.35}} = 807 \text{ MPa}$$

Material Properties and Allowable Stresses (Gear 2)

9. **Hardness:**

$$HB_p = 450$$

10. **Allowable Bending Stress (No Factors):**

$$S_{pb} = (0.533 \cdot HB_p + 88.3) \cdot 10^6 = 328 \text{ MPa}$$

11. **Allowable Contact Stress (No Factors):**

$$S_{pc} = (2.22 \cdot HB_p + 200) \cdot 10^6 = 1199 \text{ MPa}$$

12. **Geometry Factor for Bending:** Equation determined from Figure 5-46 in e-book.

$$Y_{np} = 1.6831 \cdot (N_{c2}^{-0.0323}) = 0.86$$

13. **Geometry Factor for Contact:** Equation determined from Figure 5-46 in e-book.

$$Z_{np} = 2.466 \cdot (N_{c2}^{-0.056}) = 0.78$$

14. **Load Sharing Factor:**

$$C_{hg} = 1 + ((8.89 \cdot 10^{-3}) \cdot \frac{HB_p}{H_{G_G}} - 8.29 \cdot 10^{-3}) \left(\frac{N_G}{N_P} - 1 \right) = 0.998$$

15. **Allowable Bending Stress (With Factors):**

$$S_{all,bp} = \frac{S_{pb} \cdot Y_{np}}{K_t \cdot K_r} = \frac{328 \cdot 0.86}{1 \cdot 1} = 283 \text{ MPa}$$

16. **Allowable Contact Stress (With Factors):**

$$S_{all,cp} = \frac{S_{pc} \cdot Z_{np} \cdot C_{hp}}{K_t \cdot K_r} = \frac{1199 \cdot 0.78 \cdot 0.998}{1 \cdot 1} = 928 \text{ MPa}$$

Factors of Safety

- **Bending Stress Factor of Safety (n_{bp}):**

$$n_{bp} = \frac{S_{all,bp}}{\sigma_{pb}} = \frac{283}{62.2} = 4.56$$

- **Contact Stress Factor of Safety (n_{cp}):**

$$n_{cp} = \frac{S_{all,cp}}{\sigma_{pc}} = \frac{1199}{599} = 1.55$$

Material Properties and Allowable Stresses (Gear 3)

17. **Hardness:**

$$HB_p = 578$$

18. **Allowable Bending Stress (No Factors):**

$$S_{pb} = (0.533 \cdot HB_p + 88.3) \cdot 10^6 = 396 \text{ MPa}$$

19. **Allowable Contact Stress (No Factors):**

$$S_{pc} = (2.22 \cdot HB_p + 200) \cdot 10^6 = 1483 \text{ MPa}$$

20. **Geometry Factor for Bending:** Equation determined from Figure 5-46 in e-book.

$$Y_{np} = 1.6831 \cdot (N_{c3}^{-0.0323}) = 0.86$$

21. **Geometry Factor for Contact:** Equation determined from Figure 5-46 in e-book.

$$Z_{np} = 2.466 \cdot (N_{c3}^{-0.056}) = 0.78$$

22. **Load Sharing Factor:**

$$C_{hg} = 1 + ((8.89 \cdot 10^{-3}) \cdot \frac{HB_P}{HG_G} - 8.29 \cdot 10^{-3}) \left(\frac{N_G}{N_P} - 1 \right) = 1.005$$

23. **Allowable Bending Stress (With Factors):**

$$S_{all,bp} = \frac{S_{pb} \cdot Y_{np}}{K_t \cdot K_r} = \frac{396 \cdot 0.86}{1 \cdot 1} = 342 \text{ MPa}$$

24. **Allowable Contact Stress (With Factors):**

$$S_{all,cp} = \frac{S_{pc} \cdot Z_{np} \cdot C_{hp}}{K_t \cdot K_r} = \frac{1483 \cdot 0.78 \cdot 1.005}{1 \cdot 1} = 1157 \text{ MPa}$$

Factors of Safety

- **Bending Stress Factor of Safety (n_{bp}):**

$$n_{bp} = \frac{S_{all,bp}}{\sigma_{pb}} = \frac{342}{140} = 2.45$$

- **Contact Stress Factor of Safety (n_{cp}):**

$$n_{cp} = \frac{S_{all,cp}}{\sigma_{pc}} = \frac{1157}{807} = 1.43$$

Shaft 3 Gear Analysis:

Key Parameters

Parameter	Symbol	Equation	Value	Unit
Train Value	i	$i = 6.75$	6.75	-
Input Power	H	$H = 60,000$	60,000	W
Torque	T_4	$T_4 = T_1 \cdot i$	1432.60	Nm
Number of Teeth (Gear 1)	N_4	# of teeth	61	-
Addendum	a	$a = m$	5	mm
Dedendum	b	$b = 1.25m$	6	mm
Tooth Height	h	$h = a + b$	11	mm
Clearance	c	$c = b - a$	1	mm
Circular Pitch	p_c	$p_c = \pi m$	16	mm
Base Pitch	p_b	$p_b = \pi m \cos(\phi)$	15	mm
Base Width	b_w	$b_w = 13m$	65	mm
Tooth Thickness	t	$t = 8$	8	mm
Pitch Circle Diameter	d_{p4}	$d_{p4} = N_4 \cdot m$	305	mm
Outside Circle Diameter	d_{o4}	$d_{o4} = d_{p4} + 2a$	315	mm
Base Circle Diameter	d_{b4}	$d_{b4} = d_{p4} \cos(\phi)$	287	mm
Root Circle Diameter	d_{r4}	$d_{r4} = d_{p4} - 2b$	292.5	mm
Tangential Force	W_{t4}	$W_{t4} = \frac{T_4}{d_{p4}/2}$	9394.13	N
Velocity	V_{34}	$V_{34} = \frac{\pi n_4 d_{p4}}{60}$	6.39	m/s
Operation Hours	t	$t = 5 \text{ years} \times 52 \times 7 \times 8$	14,560	hrs
Number of Cycles	N_{c4}	$N_{c4} = \frac{60 t n_4}{10^6}$	349.39	Mc

Equations and Calculations

1. **Radial Force (W_r):** The radial force is calculated using the pressure angle:

$$W_r = W_t \cdot \tan(\phi)$$

Substituting values:

$$W_r = 9394.18 \cdot \tan(20^\circ) = 3419.18N$$

2. **Normal Force (W_n):** The normal force is derived from the tangential force:

$$W_n = \frac{W_t}{\cos(\phi)}$$

Substituting values:

$$W_n = \frac{9394.18}{\cos(20^\circ)} = 9997.02 N$$

Stress Analysis

7. **Bending Stress (σ_{pb}):** The bending stress is calculated using:

$$\sigma_{gb} = \frac{W_{t4} \cdot k_b}{b_w \cdot m \cdot Y_j}$$

Where:

- o Bending geometry factor: $Y_j = 0.41$ (from Figure 5-50 in e-book)
- o Load factor:

$$k_b = K_b \cdot K_i \cdot K_a \cdot K_m \cdot K_v$$

$$\blacksquare K_b = 1, K_i = 1, K_a = 1, K_m = 1.2, K_v = \frac{59.7 + (200V_{34})^{0.5}}{59.7}^{0.825}$$

Substituting Velocity $V_{l2} = 6.39 m/s$:

$$K_v = \left(\frac{59.7 + \sqrt{200 \cdot 6.39}}{59.7} \right)^{0.825} = 1.47$$

$$k_b = 1 \cdot 1 \cdot 1 \cdot 1.2 \cdot 1.47 = 1.77$$

Substituting into the stress equation:

$$\sigma_{pb} = \frac{9394.13 \cdot 1.77}{60 \cdot 5 \cdot 0.41} = 135.03 MPa$$

8. **Contact Stress (σ_{pc}):** The contact stress is given by:

$$\sigma_{gc} = K_e \cdot \sqrt{\frac{W_t \cdot k_c}{b_w \cdot d_{pl} \cdot I}}$$

Where:

$$K_e = \sqrt{\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2}}$$

With material properties:

$$\circ E_1 = E_2 = 2.07 \times 10^{11} Pa, \nu_1 = \nu_2 = 0.3$$

$$K_e = \sqrt{\frac{1 - 0.3^2}{2.07 \times 10^{11}} + \frac{1 - 0.3^2}{2.07 \times 10^{11}}} = 473675$$

Geometry factor:

$$I = \pi \cdot \cos(\phi) \cdot \sin(\phi) \left(1 + \frac{d_{p1}}{d_{p2}} \right)$$

Substituting:

$$I = \pi \cdot \cos(20^\circ) \cdot \sin(20^\circ) \cdot \left(1 + \frac{110}{280} \right) = 0.72$$

Load factor:

$$k_c = 2.07$$

Substituting into the contact stress equation:

$$\sigma_{pc} = 473675 \cdot \sqrt{\frac{3858 \cdot 2.07}{65 \cdot 110 \cdot 0.72}} = 858.24 \text{ MPa}$$

Material Properties and Allowable Stresses

Hardness:

$$HB_g = 450$$

Allowable Bending Stress (No Factors):

$$S_{gb} = (0.533 \cdot HB_g + 88.3) \cdot 10^6 = 328.15 \text{ MPa}$$

Allowable Contact Stress (No Factors):

$$S_{gc} = (2.22 \cdot HB_g + 200) \cdot 10^6 = 1199 \text{ MPa}$$

Geometry Factor for Bending: Equation determined from Figure 5-46 in e-book.

$$Y_{ng} = 1.6831 \cdot (N_{c4}^{-0.0323}) = 0.89$$

Geometry Factor for Contact: Equation determined from Figure 5-46 in e-book.

$$Z_{ng} = 2.466 \cdot (N_{c4}^{-0.056}) = 0.82$$

Allowable Bending Stress (With Factors):

$$S_{all,bg} = \frac{S_{pb} \cdot Y_{np}}{K_t \cdot K_r} = \frac{519 \cdot 0.84}{1 \cdot 1} = 292.57 \text{ MPa}$$

Allowable Contact Stress (With Factors):

$$S_{all,cg} = \frac{S_{pc} \cdot Z_{np} \cdot C_{hp}}{K_t \cdot K_r} = \frac{1629 \cdot 0.74 \cdot 1}{1 \cdot 1} = 987.70 \text{ MPa}$$

Factors of Safety

1. **Bending Stress Factor of Safety (n_{bp}):**

$$n_{bp} = \frac{S_{all,bp}}{\sigma_{pb}} = \frac{292.57}{135.03} = 2.16$$

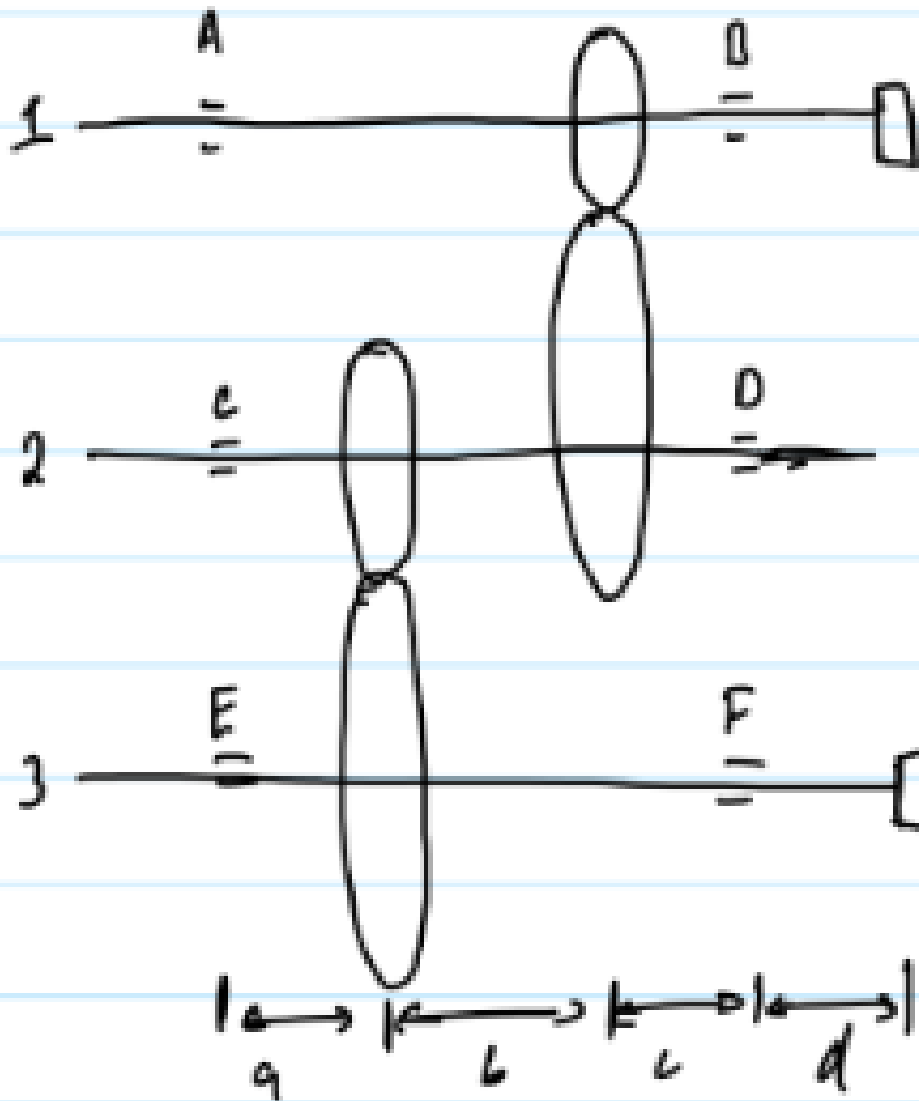
2. **Contact Stress Factor of Safety (n_{cp}):**

$$n_{cp} = \frac{S_{all,cp}}{\sigma_{pc}} = \frac{987.70}{858.24} = 1.15$$

Appendix 3 - Bearing Selection

Bearing Label Schematic:

Actual:



Shaft 1 Bearing Selection:

Key Parameters

Parameter	Symbol	Equation	Value	Unit
Radial Force at A	F_{rA}	$F_{rA} = \sqrt{A_y^2 + A_z^2}$	1171	N
Radial Force at B	F_{rB}	$F_{rB} = \sqrt{B_y^2 + B_z^2}$	3514	N
Life in Millions of Cycles	L_{10}	From shaft analysis	2358.72	Mc
Application Factor	A_p	$A_p = 1$ (No additional load factors)	1	-
Bearing Type	-	SKF Deep Groove Ball Bearing	-	-

Bearing Selection

1. For Bearing at B:

- o Load on the bearing:

$$P = A_p \cdot F_{rB} = 1 \cdot 3514 = 3514 \text{ N}$$

- o Dynamic load capacity:

$$C = P \cdot L_{10}^{1/m}$$

Substituting $m = 3$:

$$C = 3514 \cdot 2358.72^{1/3} = 46780 \text{ N}$$

- o Selected Bearing:

Bearing 6309: $d_b = 45 \text{ mm}$, $OD = 100 \text{ mm}$, $C = 52700 \text{ N}$, $w = 25 \text{ mm}$

2. For Bearing at A:

- o Load on the bearing:

$$P = A_p \cdot F_{rA} = 1 \cdot 1171 = 1171 \text{ N}$$

- o Dynamic load capacity:

$$C = P \cdot L_{10}^{1/m}$$

Substituting $m = 3$:

$$C = 1171 \cdot 2358.72^{1/3} = 15583 \text{ N}$$

- o Selected Bearing:

Bearing 6304: $d_b = 20 \text{ mm}$, $OD = 52 \text{ mm}$, $C = 15000 \text{ N}$, $w = 15 \text{ mm}$

Final Bearing Selection

Bearing 6309: (same for both): $d_b = 45 \text{ mm}$, $OD = 100 \text{ mm}$, $C = 52700 \text{ N}$, $w = 25 \text{ mm}$

No bearing changes are required during the five year service period.

Shaft 2 Bearing Selection:

Key Parameters

Parameter	Symbol	Equation	Value	Unit
Radial Force at C	F_{rA}	$F_{rA} = \sqrt{A_y^2 + A_z^2}$	8105	N
Radial Force at D	F_{rB}	$F_{rB} = \sqrt{B_y^2 + B_z^2}$	5998	N
Life in Millions of Cycles	L_{10}	From shaft analysis	926.64	Mc
Application Factor	A_p	$A_p = 1$ (No additional load factors)	1	-
Bearing Type	-	SKF Deep Groove Ball Bearing	-	-

Bearing Selection

3. For Bearing at C:

- o Load on the bearing:

$$P = A_p \cdot F_{rB} = 1 \cdot 8105 = 8105 \text{ N}$$

- o Dynamic load capacity:

$$C = P \cdot L_{10}^{1/m}$$

Substituting $m = 3$:

$$C = 8105 \cdot 926.64^{1/3} = 79016 \text{ N}$$

- o Selected Bearing:

Bearing 6410: $d_b = 55 \text{ mm}$, $OD = 130 \text{ mm}$, $C = 87100 \text{ N}$, $w = 31 \text{ mm}$

4. For Bearing at D:

- o Load on the bearing:

$$P = A_p \cdot F_{rA} = 1 \cdot 5998 = 5998 \text{ N}$$

- o Dynamic load capacity:

$$C = P \cdot L_{10}^{1/m}$$

Substituting $m = 3$:

$$C = 5998 \cdot 926.65 = 58477 \text{ N}$$

- o Selected Bearing:

Bearing 6310: $d_b = 55 \text{ mm}$, $OD = 110 \text{ mm}$, $C = 61800 \text{ N}$, $w = 27 \text{ mm}$

Final Bearing Selection

Bearing 6410: (same for both): $d_b = 55 \text{ mm}$, $OD = 130 \text{ mm}$, $C = 87100 \text{ N}$, $w = 31 \text{ mm}$

No bearing changes are required during the five year service period.

Shaft 3 Bearing Selection:

Key Parameters

Parameter	Symbol	Equation	Value	Unit
Radial Force at E	F_{rA}	$F_{rA} = \sqrt{A_y^2 + A_z^2}$	6511.88	N
Radial Force at F	F_{rB}	$F_{rB} = \sqrt{B_y^2 + B_z^2}$	7394.23	N
Life in Millions of Cycles	L_{10}	From shaft analysis	349.39	Mc
Application Factor	A_p	$A_p = 1$ (No additional load factors)	1	-
Bearing Type	-	SKF Deep Groove Ball Bearing	-	-

Bearing Selection

5. For Bearing at E:

- o Load on the bearing:

$$P = A_p \cdot F_{rB} = 1 \cdot 6511.88 = 6511.88 \text{ N}$$

- o Dynamic load capacity:

$$C = P \cdot L_{10}^{1/m}$$

Substituting $m = 3$:

$$C = 6511.88 \cdot 349.39^{1/3} = 45864.43 \text{ N}$$

- o Selected Bearing:

Bearing 6209: $d_b = 45 \text{ mm}$, $OD = 85 \text{ mm}$, $C = 33200 \text{ N}$, $w = 19 \text{ mm}$

6. For Bearing at G:

- o Load on the bearing:

$$P = A_p \cdot F_{rA} = 1 \cdot 7394.23 = 7394.23 \text{ N}$$

- o Dynamic load capacity:

$$C = P \cdot L_{10}^{1/m}$$

Substituting $m = 3$:

$$C = 7394.23 \cdot 349.39^{1/3} = 52079.02 \text{ N}$$

- o Selected Bearing:

Bearing 6309: $d_b = 45 \text{ mm}$, $OD = 100 \text{ mm}$, $C = 52700 \text{ N}$, $w = 25 \text{ mm}$

Final Bearing Selection

Bearing 6309: (same for both): $d_b = 45 \text{ mm}$, $OD = 100 \text{ mm}$, $C = 52700 \text{ N}$, $w = 25 \text{ mm}$

No bearing changes are required during the five year service period.